



Cochrane
Library

Cochrane Database of Systematic Reviews

Occupational therapy for multiple sclerosis (Review)

Kos D, Boers A, O'Meara C, Bekkering GE, De Coninck L, Koen M, Freeman J, Hynes SM, Eijssen ICJM

Kos D, Boers A, O'Meara C, Bekkering GE, De Coninck L, Koen M, Freeman J, Hynes SM, Eijssen ICJM.
Occupational therapy for multiple sclerosis.
Cochrane Database of Systematic Reviews 2026, Issue 1. Art. No.: CD015371.
DOI: [10.1002/14651858.CD015371.pub2](https://doi.org/10.1002/14651858.CD015371.pub2).

www.cochranelibrary.com

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	13
OBJECTIVES	14
METHODS	14
Figure 1.	15
RESULTS	21
Figure 2.	22
DISCUSSION	27
AUTHORS' CONCLUSIONS	30
SUPPLEMENTARY MATERIALS	31
ADDITIONAL INFORMATION	31
REFERENCES	34
ADDITIONAL TABLES	43

[Intervention Review]

Occupational therapy for multiple sclerosis

Daphne Kos^{1,2a}, Ashley Boers^{1a}, Ciara O'Meara³, Geertruida E Bekkering^{4,5b}, Leen De Coninck^{1,6c}, Marja Koen⁷, Jennifer Freeman^{8d}, Sinéad M Hynes^{9,10e}, Isaline CJM Eijssen^{11,12,13,14f}

¹KU Leuven, Department of Rehabilitation Sciences, Research Group for Neurorehabilitation, Leuven, Belgium. ²National Multiple Sclerosis Center, Melsbroek, Belgium. ³School of Nursing & Midwifery, University of Galway, Galway, Ireland. ⁴Belgian Centre for Evidence-Based Medicine, Cochrane Belgium, Leuven, Belgium. ⁵KU Leuven, Department of Public Health and Primary Care, Academic Centre for General Practice, Leuven, Belgium. ⁶Working Group Development of Primary Care Guidelines, Worel, Antwerp, Belgium. ⁷MS Telefoon, Dutch MS Association, Den Dolder, Netherlands. ⁸School of Health Professions, Faculty of Health, University of Plymouth, Plymouth, UK. ⁹Discipline of Occupational Therapy, School of Health Sciences, College of Medicine, Nursing and Health Sciences, University of Galway, Galway, Ireland. ¹⁰Galway Neuroscience Centre, Galway, Ireland. ¹¹Amsterdam UMC location Vrije Universiteit Amsterdam, Rehabilitation Medicine, MS Center Amsterdam, Amsterdam, Netherlands. ¹²Amsterdam Neuroscience, Neuroinfection & -inflammation, Amsterdam, Netherlands. ¹³Amsterdam Public Health, Social Participation & Health, Amsterdam, Netherlands. ¹⁴Amsterdam Movement Sciences, Rehabilitation & Development, Amsterdam, Netherlands

^aThese authors should be considered joint first author ORCID iD: <https://orcid.org/0000-0003-3566-2269>; ORCID iD: <https://orcid.org/0009-0008-6574-3696>. ^bORCID iD: <https://orcid.org/0000-0002-3845-1507>. ^cORCID iD: <https://orcid.org/0000-0002-1060-2422>. ^dORCID iD: <https://orcid.org/0000-0002-4072-9758>. ^eORCID iD: <https://orcid.org/0000-0002-3199-7355>. ^fORCID iD: <https://orcid.org/0009-0000-3113-8676>

Contact: Daphne Kos, daphne.kos@kuleuven.be.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 1, 2026.

Citation: Kos D, Boers A, O'Meara C, Bekkering GE, De Coninck L, Koen M, Freeman J, Hynes SM, Eijssen ICJM. Occupational therapy for multiple sclerosis. *Cochrane Database of Systematic Reviews* 2026, Issue 1. Art. No.: CD015371. DOI: [10.1002/14651858.CD015371.pub2](https://doi.org/10.1002/14651858.CD015371.pub2).

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Rationale

Occupational therapy (OT) enables individuals to engage in meaningful daily activities and is considered a valuable component of care for people with multiple sclerosis (MS). However, the specific impact of OT on MS remains unclear.

Objectives

To assess the benefits and harms of occupational therapy interventions for improving daily functioning, participation and quality of life in people with multiple sclerosis.

Search methods

We searched seven electronic bibliographic databases until November 2024. We also searched grey literature and trial registers.

Eligibility criteria

We included randomised and non-randomised controlled trials that compared OT for adults with MS versus no intervention, usual care or active control.

Outcomes

Critical outcomes were daily functioning, quality of life and adverse effects. Important outcomes were participation, self-efficacy, self-management, mood, resilience and impact on caregivers. Our time points of interest were post-intervention, medium term (up to six months after the end of the intervention) and long term (longest follow-up after six months).

Occupational therapy for multiple sclerosis (Review)

Copyright © 2026 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

Risk of bias

We assessed risk of bias (RoB) for the outcomes reported in our summary of findings tables. We used the Cochrane tools RoB-2 (for randomised controlled trials) and ROBINS-I (for non-randomised controlled trials).

Synthesis methods

We synthesised results using meta-analysis (random-effects model, inverse variance). The effect measure was mean difference (MD) or standardised mean difference (SMD). Where meta-analysis was not possible, we synthesised results narratively using Synthesis Without Meta-analysis guidelines. We used GRADE to assess the certainty of the evidence.

Included studies

We included 20 studies (1628 participants), three of which were non-randomised. Nineteen of the studies were conducted in high-income countries. The studies included adults with MS (aged 18 to 70 years) with low to moderate levels of disability. Ten studies evaluated fatigue management programmes, nine examined OT interventions for daily functioning (e.g. skills training) and one targeted social participation.

Synthesis of results

Daily functioning

Post-intervention: OT interventions may provide a small benefit for daily functioning compared to active control (SMD 0.22, 95% CI -0.02 to 0.46; $I^2 = 17\%$; 7 studies, 364 participants; low certainty), and a moderate benefit compared to no intervention (SMD 0.56, 95% CI -0.26 to 1.37; $I^2 = 68\%$; 2 studies, 164 participants; low certainty). There may be a large effect compared to usual care, but the evidence is very uncertain (SMD 1.19, 95% CI 0.29 to 2.09; $I^2 = 62\%$; 2 studies, 120 participants; very low certainty).

Medium term: OT may make little to no difference to daily functioning compared to active control (SMD 0.11, 95% CI -0.13 to 0.35; $I^2 = 0\%$; 4 studies, 264 participants; low certainty), but may have a large effect compared to usual care (SMD 0.99, 95% CI 0.57 to 1.41; I^2 not applicable; 1 study, 98 participants; low certainty).

The long-term effects of OT on daily functioning compared to active control or usual care are very uncertain.

No medium-term or long-term data were reported for OT versus no intervention.

Quality of life

Post-intervention: OT may offer a small benefit for mental health-related quality of life (HR-QoL) versus active control (SMD 0.33, 95% CI -0.01 to 0.66; $I^2 = 42\%$; 5 studies, 296 participants; low certainty), but little to no benefit for physical HR-QoL (SMD 0.17, 95% CI -0.09 to 0.42; $I^2 = 12\%$; 5 studies, 295 participants; low certainty). Compared to usual care, moderate effects on physical HR-QoL (SMD 0.69, 95% CI -1.18 to 2.56; $I^2 = 91\%$; 2 studies, 166 participants; very low certainty) and mental HR-QoL (SMD 0.44, 95% CI -1.27 to 2.16; $I^2 = 91\%$; 2 studies; very low certainty) are very uncertain. Compared to no intervention, OT may moderately improve mental HR-QoL (SMD 0.68, 95% CI -0.09 to 1.45; $I^2 = 0\%$; 3 studies, 192 participants; very low certainty), though the evidence is very uncertain; there may be little to no effect on physical HR-QoL (SMD 0.12, 95% CI -0.62 to 0.86; $I^2 = 0\%$; 3 studies, 192 participants; very low certainty), but the evidence is very uncertain.

Medium term: there may be a small benefit from OT on mental HR-QoL versus active control, but the evidence is very uncertain (SMD 0.21, 95% CI -0.08 to 0.49; $I^2 = 20\%$; 4 studies, 270 participants; very low certainty). There may be little to no effect on physical HR-QoL (SMD 0.19, 95% CI -0.05 to 0.43; $I^2 = 0\%$; 4 studies, 268 participants; low certainty). Compared to usual care, effects are very uncertain for both physical (SMD 0.26, 95% CI -0.33 to 0.86; $I^2 = 81\%$; 2 studies, 242 participants; very low certainty) and mental HR-QoL (SMD 0.18, 95% CI -0.63 to 0.98; $I^2 = 90\%$; 2 studies, 242 participants; very low certainty). No results are available for medium-term effects of OT compared to no intervention.

The long-term effects of OT on quality of life are very uncertain.

Adverse effects

No studies systematically reported the frequency or type of adverse effects.

Participation

Occupational therapy may result in little to no effect on participation post-intervention compared to active control. Results for other comparisons and time points are not available or provide only evidence of very low certainty.

Authors' conclusions

Occupational therapy may improve daily functioning and mental-health-related quality of life post-intervention, regardless of the comparator intervention. These potential benefits may persist in the medium term for daily functioning (when compared to usual care) and for mental-health-related quality of life (when compared to active control). Long-term effects are uncertain.

Sparse evidence suggests that occupational therapy may have little or no effect on physical-health-related quality of life compared to active comparators.

We found no evidence about adverse effects of occupational therapy.

Our pooled effects require cautious interpretation, as intervention heterogeneity and our exclusion of multidisciplinary studies without separate data for occupational therapy limit our certainty about the evidence.

Future studies with robust design and systematic outcome assessment are needed to reach firm conclusions about the effects of occupational therapy for people with multiple sclerosis.

Funding

Elizabeth Casson Trust Research Grant_2022 (UK); Health Research Board (Ireland) & HSC Public Health Agency (Grant ESI-2021-001) through Evidence Synthesis/Cochrane Ireland

Registration

Protocol DOI 10.1002/14651858.CD015371

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of occupational therapy for people with multiple sclerosis?

Key messages

- Immediately after occupational therapy (OT), people with multiple sclerosis may show small improvements in daily functioning and mental health-related quality of life (HR-QoL), but OT may have little to no effect on physical HR-QoL or participation.
- Three to six months after therapy, OT may lead to small improvements in mental HR-QoL, but may have little effect on daily activities or physical HR-QoL. Compared to usual care, occupational therapy may cause some improvement in daily functioning and physical and mental health-related QoL.
- Due to lack of information, we are uncertain about the long-term benefits or harms of OT.
- The real effect of OT may be undervalued in our findings because the studies tested a wide range of different OT interventions, and we did not include studies that evaluated a combination of OT and other therapies unless the specific effects of the OT were measured.
- Future research should aim to identify which components work best for whom and in what context.

What is multiple sclerosis?

Multiple sclerosis (MS) is a common disease of the central nervous system, affecting about 2.9 million people worldwide. In MS, the immune system mistakenly attacks the protective layer around nerve fibres in the brain and spinal cord. This disrupts communication between the brain and the rest of the body, causing symptoms like fatigue, muscle weakness and balance problems. MS can make everyday tasks like dressing, preparing meals or walking difficult. It can also affect memory and concentration, making work and social interactions more challenging.

How is multiple sclerosis treated?

Medications for MS aim to reduce flare-ups, slow disease progression and manage symptoms. Rehabilitation therapies can support people with MS to manage their symptoms, improve daily functioning and stay active in the community.

One rehabilitation therapy is occupational therapy (OT). OT helps people with everyday tasks like dressing, cooking, bathing and working. Occupational therapists teach people ways to make these activities easier and provide tools to help manage symptoms. They also work on improving strength and coordination to make daily life more manageable.

What did we want to find out?

We wanted to find out how OT benefits people with MS, specifically how it affects daily functioning, physical and mental health-related quality of life (HR-QoL), and participation. We also wanted to find out if OT causes any unwanted effects.

What did we do?

We searched for studies that compared OT interventions with other interventions, usual care or no treatment in people with MS. We excluded multidisciplinary studies — studies where several different types of health professionals provide care in parallel — unless the contribution of OT was specifically reported. We were interested in the effects immediately after treatment, three to six months after treatment (medium term) and one year after treatment (long term). We combined the results of the studies and rated our confidence in the evidence, based on factors such as study methods and number of studies.

What did we find?

We found 20 studies of OT interventions that involved 1628 people with MS. Ten studies looked at ways to help people with MS manage their fatigue; nine focused on improving daily activities like dressing and cooking; and one aimed to support greater involvement in social activities. The studies were mostly carried out in high-income countries. The participants in the studies were adults aged 18 to 70 years who had MS causing mild to moderate disability.

What were the results?

OT may give small improvements in daily functioning compared with other treatments in the short term, and possibly larger benefits compared with usual care or no intervention, but the evidence is often uncertain. Medium-term and long-term effects are unclear.

OT may lead to small improvements in mental HR-QoL in the short term, but have little to no effect on physical HR-QoL. Results for comparisons with usual care or no intervention are very uncertain, and medium-term and long-term effects are unclear.

OT may have little to no effect on participation in the short term; evidence is very limited.

No study reported harms from OT in a systematic way, so these remain unknown.

What are the limitations of the evidence?

There is limited evidence about how OT affects physical HR-QoL for people with MS. We are unsure how OT compares to usual care or no treatment because there are not enough studies to draw firm conclusions. There is no evidence about potential harmful effects of OT, and limited evidence about OT for people with high levels of disability due to MS.

Another limitation was that we grouped very different OT interventions together, making the results less certain and harder to interpret. OT is made up of many different approaches, not just one. Also, by leaving out most team-based studies unless the role of OT was clearly described, we may have missed important evidence.

More studies are needed to fully understand the impact of OT on people with MS.

How up to date is this evidence?

The evidence is based on searches run in November 2024.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Occupational therapy compared to active intervention for improvement post-intervention in people with multiple sclerosis

Occupational therapy compared to active intervention for improvement post-intervention in people with multiple sclerosis

Patient or population: improvement post-intervention in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: active intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with active intervention	Risk with occupational therapy				
Daily functioning - post-intervention	-	SMD 0.22 SD higher (0.02 lower to 0.46 higher)	-	379 (7 RCTs)	⊕⊕⊕⊕ Low ^{a,b,c}	Occupational therapy may result in a small benefit on daily functioning compared with an active intervention in the short term.
Quality of life - physical health - post-intervention	-	SMD 0.17 higher (0.09 lower to 0.42 higher)	-	302 (5 RCTs)	⊕⊕⊕⊕ Low ^{b,c,d}	Occupational therapy may result in little to no effect on quality of life - physical health compared with an active intervention in the short term.
Quality of life - mental health - post-intervention	-	SMD 0.33 higher (0.01 lower to 0.66 higher)	-	302 (5 RCTs)	⊕⊕⊕⊕ Low ^{b,c,d}	Occupational therapy may result in a small benefit on quality of life - mental health compared with an active intervention in the short term.
Adverse effects - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - post-intervention	-	SMD 0.18 higher (0.18 lower to 0.53 higher)	-	215 (2 RCTs)	⊕⊕⊕⊕ Low ^{b,c,d}	Occupational therapy may result in little to no effect for participation compared with an active intervention in the short term.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_456262126874433432.

^a No serious study limitations; although we have some concerns with respect to unblinding in 4 studies examining fatigue management that used self-reported fatigue, we judge overall risk of bias not serious enough to warrant downgrading. For remaining studies we have no concerns.

^b Serious indirectness (-1); studies examine specific interventions within occupational therapy, for example 4 studies examine fatigue management interventions, which is considered to contribute only partly to the more general outcomes daily functioning, quality of life or participation.

^c Serious imprecision (-1); the confidence interval includes contradictory conclusions (no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

^d No serious study limitations; although we have some concerns regarding unblinding in the studies, we judge the overall risk of bias not serious enough to warrant downgrading.

Summary of findings 2. Summary of findings table - Occupational therapy compared to active intervention for improvement in the medium term in people with multiple sclerosis

Occupational therapy compared to active intervention for improvement in the medium term in people with multiple sclerosis

Patient or population: improvement in the medium term in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: active intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with active intervention	Risk with occupational therapy				
Daily functioning - medium term	-	SMD 0.11 higher (0.13 lower to 0.35 higher)	-	281 (4 RCTs)	⊕⊕⊕⊕ Low ^{a,b,c}	Occupational therapy may result in little to no effect on daily functioning compared with an active intervention in the medium term.
Quality of life - physical health - medium-term	-	SMD 0.19 higher (0.05 lower to 0.43 higher)	-	281 (4 RCTs)	⊕⊕⊕⊕ Low ^{b,c,d}	Occupational therapy may result in little to no effect on quality of life - physical health compared with an active control in the medium term.
Quality of life - mental health - medium term	-	SMD 0.21 SD higher (0.08 lower to 0.49 higher)	-	281 (4 RCTs)	⊕⊕⊕⊕ Very low ^{b,d,e}	Occupational therapy may result in a small benefit on quality of life - mental health compared with an active control in the medium term, but the evidence is very uncertain.

Adverse effects - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - medium term	-	SMD 0.02 higher (0.26 lower to 0.29 higher)	-	215 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,d,e}	Occupational therapy may result in little to no effect on participation compared with an active control in the medium term, but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_456262241289541714.

^a No serious study limitations; although we have some concerns with respect to unblinding of outcomes in 4 studies examining fatigue management studies and that used self-reported fatigue, we judge overall risk of bias not serious enough to warrant downgrading. For remaining studies we have no concerns.

^b Serious indirectness (-1): studies examine specific interventions within occupational therapy, for example studies examine fatigue management interventions, which is considered to contribute only partly to the more general outcomes daily functioning, quality of life or participation.

^c Serious imprecision (-1); the confidence interval includes contradictory conclusions (no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

^d No serious study limitations; although we have some concerns regarding unblinding in the studies, we judge the overall risk of bias not serious enough to warrant downgrading.

^e Very serious imprecision (-2); the confidence interval includes 3 contradictory conclusions (appreciable harm, no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

Summary of findings 3. Summary of findings table - Occupational therapy compared to usual care for improvement post-intervention in people with multiple sclerosis

Occupational therapy compared to usual care for improvement post-intervention in people with multiple sclerosis

Patient or population: improvement post-intervention in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: usual care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with usual care	Risk with occupational therapy				
Daily functioning - post-intervention	-	SMD 1.19 higher (0.29 higher to 2.09 higher)	-	130 (2 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	Occupational therapy may have a large effect on daily functioning compared to usual care in the short term, but the evidence is very uncertain.
Quality of life - physical health - post-intervention	-	SMD 0.69 higher (1.18 lower to 2.56 higher)	-	184 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,d,e}	Occupational therapy may have a moderate effect on quality of life – physical health compared to usual care in the short term, but the evidence is very uncertain.
Quality of life - mental health - post-intervention	-	SMD 0.44 higher (1.27 lower to 2.16 higher)	-	184 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,d,e}	Occupational therapy may have a small effect on quality of life – mental health compared to usual care in the short term, but the evidence is very uncertain.
Adverse effects - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - post-intervention - not reported	-	-	-	-	-	No results were available for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457183469642571097.

^a Serious study limitations (-1); due to bias in measurement of the outcome.

^b Serious indirectness (-1); studies examined specific components of occupational therapy, only partially addressing broader outcomes like daily functioning, quality of life, and participation.

^c Very serious imprecision (-2); the confidence interval include contradictory conclusion (appreciable harm, no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

^d Serious study limitations (-1); due to some concerns in 4 out of 5 domains of risk of bias in both studies.

^e Serious inconsistency (due to conflicting effect directions between studies and substantial unexplained heterogeneity ($I^2 = 91\%$)).

Summary of findings 4. Summary of findings table - Occupational therapy compared to usual care for improvement in the medium term in people with multiple sclerosis

Occupational therapy compared to usual care for improvement in the medium term in people with multiple sclerosis

Patient or population: improvement in the medium term in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: usual care

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with usual care	Risk with occupational therapy				
Daily functioning - medium term	-	SMD 0.99 higher (0.57 higher to 1.41 higher)	-	110 (1 RCT)	⊕⊕⊕⊕ Low ^{a,b}	Occupational therapy may result in a large improvement of daily functioning compared to usual care in the medium term.
Quality of life - physical health - medium term	-	SMD 0.26 higher (0.33 lower to 0.86 higher)	-	274 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,d,e}	Occupational therapy may result in a small benefit for quality of life - physical health compared with usual care in the medium term, but the evidence is very uncertain.
Quality of life - mental health - medium term	-	SMD 0.18 higher (0.63 lower to 0.98 higher)	-	274 (2 RCTs)	⊕⊕⊕⊕ Very low ^{b,c,d,e}	Occupational therapy may have little to no effect for quality of life - mental health compared with usual care in the medium term, but the evidence is very uncertain.
Adverse effects - not reported - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - medium term - not measured	-	-	-	-	-	No results were available for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_457183874484657707.

^a Serious study limitations (-1); due to bias in measurement of the outcome.

^b Serious indirectness (-1); studies examine specific interventions within occupational therapy, which is considered to contribute only partly to the more general outcomes daily functioning or quality of life.

^c No serious study limitations; although we have some concerns with respect to unblinding, we judge overall risk of bias not serious enough to warrant downgrading.

^d Serious inconsistency (-1) due to conflicting effect directions and substantial unexplained heterogeneity.

^e Serious imprecision (-1); the confidence interval include contradictory conclusion (no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

Summary of findings 5. Summary of findings table - Occupational therapy compared to no intervention for improvement post-intervention in people with multiple sclerosis

Occupational therapy compared to no intervention for improvement post-intervention in people with multiple sclerosis

Patient or population: improvement post-intervention in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: no intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no intervention	Risk with occupational therapy				
Daily functioning - post-intervention	-	SMD 0.56 SD higher (0.26 lower to 1.37 higher)	-	211 (2 RCTs)	⊕⊕⊕⊕ Low ^{a,b,c}	Occupational therapy may have a moderate effect on daily functioning compared with no intervention in the short term.
Quality of life - physical health - post-intervention	-	SMD 0.12 higher (0.62 lower to 0.86 higher)	-	312 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,d}	Occupational therapy may have little to no effect on quality of life – physical health compared with no intervention in the short term, but the evidence is very uncertain.

Quality of life - mental health - post-intervention	-	SMD 0.68 higher (0.09 lower to 1.45 higher)	-	312 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,d}	Occupational therapy may result in a moderate effect on quality of life – mental health compared with no intervention in the short term, but the evidence is very uncertain.
Adverse effects - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - post-intervention	-	SMD 0.38 higher (0.45 lower to 1.21 higher)	-	30 (1 RCT)	⊕⊕⊕⊕ Very low ^{b,d,e}	Occupational therapy may result in a small benefit for participation compared with no intervention, but the evidence is very uncertain.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_457182641345910643.

^a No serious for study limitations; although we have some concerns with respect to unblinding in the studies, we judge overall risk of bias not serious enough to warrant downgrading.

^b Serious indirectness (-1); studies examine specific interventions within occupational therapy, which is considered to contribute only partly to the more general outcomes daily functioning, quality of life or participation.

^c Serious imprecision (-1); the confidence interval includes contradictory conclusions (appreciable harm, no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

^d Very serious imprecision (-2); the confidence interval includes 3 contradictory conclusions (appreciable harm, no effect and appreciable benefit). Our threshold for appreciable increase or reduction is 0.2 standard deviation.

^e Serious study limitations (-1); only 2 of 3 subscales of the outcome measure were reported (reporting bias)

Summary of findings 6. Summary of findings table - Occupational therapy compared to no intervention for improvement in the medium term in people with multiple sclerosis

Occupational therapy compared to no intervention for improvement in the medium term in people with multiple sclerosis

Patient or population: improvement in the medium term in people with multiple sclerosis

Setting: hospital, rehabilitation centre or community

Intervention: occupational therapy

Comparison: no intervention

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with no intervention	Risk with occupational therapy				
Daily functioning - medium term - not measured	-	-	-	-	-	No results were available for this outcome.
Quality of life - physical health - medium term - not measured	-	-	-	-	-	No results were available for this outcome.
Quality of life - mental health - medium term - not measured	-	-	-	-	-	No results were available for this outcome.
Adverse effects - not reported	-	-	-	-	-	No results were available for this outcome.
Participation - medium term - not measured	-	-	-	-	-	No results were available for this outcome.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **OR:** odds ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_456262399956654327.

BACKGROUND

Description of the condition

Epidemiology and disease course

Multiple sclerosis (MS) is a progressive, inflammatory, demyelinating disease of the brain and spinal cord and is the most common disabling neurological condition affecting young and middle-aged adults [1]. In 2020, an estimated 2.9 million people worldwide were living with MS (36 per 100,000), with an average annual incidence of 2.1 per 100,000 and substantial regional variation [2]. This corresponds to nearly 300 people having new diagnoses globally each day. Although the distribution by sex, age and region has remained stable, MS prevalence continues to rise [3]. The average age at diagnosis globally is 32 years, a life stage often involving family and career development [4]. Four main phenotypes are described: relapsing-remitting MS (RRMS), primary-progressive MS (PPMS), secondary-progressive MS (SPMS) and progressive-relapsing MS (PRMS) [5]. Globally, about 85% of people are diagnosed initially with RRMS and around 12% with a progressive form [5]. Although people with MS still experience higher mortality rates and reduced life expectancy compared with the general population [6, 7, 8], this gap is narrowing with improved treatments [9].

Clinical features and impact

MS can affect many functions due to demyelinating lesions in the central nervous system [10]. Common symptoms include fatigue, mobility limitations, cognitive difficulties, spasticity, and problems with speech, swallowing or bladder function [11, 12]. Depression and anxiety are also frequent across the disease course and can further reduce functioning and quality of life for both people with MS and their caregivers [13, 14, 15]. Moreover, psychological symptoms, such as depression and anxiety, may impact functioning [16, 17] and quality of life [18] in people with MS and caregivers [19, 20].

Consequences for daily life

The combined physical and psychological consequences of MS affect a broad range of daily activities, participation and routines [21, 22]. Participation refers to involvement in life situations — such as roles and activities at home, work and in the community — and reflects a person's ability to engage in meaningful social, occupational and everyday contexts [23, 24]. Because the disease often requires long-term adaptation, people with MS face ongoing challenges in maintaining independence and engagement in meaningful life roles [25]. A range of interventions is therefore needed across the course of the disease to support activity, participation and overall quality of life [26, 27, 28, 29].

Occupational therapy (OT) is a key component of this care, and people with MS often identify occupational therapists as among the most important professionals involved in their management [30, 31]. However, despite this central role, the overall impact of OT on MS-related outcomes remains unclear.

Description of the intervention and how it might work

What is occupational therapy?

Occupational therapy is a client-centred health profession that promotes health and wellbeing through supporting people to

engage in everyday activities that they need, want or are expected to do [32, 33]. Its primary goal is to enable meaningful participation in daily life.

What does an occupational therapist do?

Occupational therapists work with people to help them manage symptoms, improve activity performance and participate more fully in chosen life situations. They assess which activities an individual wants to maintain, improve or resume, and collaborate with the person to develop a tailored treatment plan. Interventions may include: training activity-specific skills (occupation-based practice), adapting activity performance (such as modifying routines or strategies), adjusting the environment (including home or workplace adaptations), education, consultation, and problem-solving, advocacy (such as supporting access to resources or assistive devices) and therapeutic use of self, to enhance engagement and adherence [34, 35].

These approaches address change at three levels: person (symptom management, coping, self-management, self-efficacy), activity (optimising performance and activity demands) and environment (adaptations, caregiver support, social networks) [34].

Where is occupational therapy provided?

OT services for people with MS are delivered across a range of settings, including:

- outpatient or day treatment in hospitals or rehabilitation centres;
- inpatient rehabilitation (in specialised rehabilitation units or hospital wards providing 24-hour care);
- community settings, such as the person's home, workplace or school [36, 37].

Because occupational therapists aim to support people in real-life contexts, delivering services in the environment where activities occur is often preferred [38]. Differences in models of care, service delivery frequency and programme structure vary across countries and settings [39, 40, 41].

What is telehealth?

Telehealth refers to the delivery of evaluative, consultative, preventive or therapeutic services through information and communication technologies [42]. It is a mode of service delivery, not a separate intervention, and can be used across all OT settings [37]. Telehealth is increasingly used in OT to reach people who are not in the same physical location as the therapist [37, 43, 44, 45].

What formats of OT exist?

OT may be delivered:

- individually, allowing therapy to be tailored to personal goals and needs;
- in groups, which can offer peer support and shared learning [46];
- in combined formats, depending on client needs and programme design.

Exploring differences in outcomes between individual and group delivery is therefore relevant.

How can OT support people with MS?

Occupational therapists may help people with MS engage in meaningful activities where they live, work and socialise. Occupational therapy can influence outcomes through several mechanisms. At the individual level, OT may strengthen physical, cognitive and social abilities, support emotional regulation, enhance resilience when dealing with challenging life situations, and promote self-management and self-efficacy by helping people make informed choices and use their own abilities [34, 47]. At the social level, OT may also affect the person's environment by supporting family members' and carers' coping and by facilitating practical, informational and emotional assistance [48].

Why it is important to do this review

In a worldwide, multi-stakeholder James Lind Alliance Priority Setting Partnership project run in 2022, two domains relevant to OT were ranked within the top five priorities for future research in multiple sclerosis [49]:

- the efficacy of multidisciplinary care by teams of different social and health professionals in improving health outcomes and experiences for people with MS; and
- the impact of psychological health on disease progression in people with MS.

The National Institute for Health and Care Excellence (NICE) recommends that occupational therapists should be included within multidisciplinary care teams as they are amongst the "professionals who can best meet the needs of the person with MS and who have expertise in managing MS" [50], page 11.

OT provides support for 'invisible symptoms' like fatigue, pain, bowel/bladder dysfunction, sexual dysfunction and vision changes [49]. OT can play an important role in managing both visible and invisible ME symptoms and helping people live fulfilling lives despite these symptoms. This was acknowledged by the James Lind Alliance (JLA) Priority Setting Partnership that prioritised research to explore the unique contribution of OT to the daily lives of service users [51].

Although the use of OT for people with MS is well-established, the evidence for its effectiveness has not been assessed through a synthesis of the existing evidence. The latest Cochrane review was carried out in 2003; and, based on the single randomised controlled trial available at that time, could not draw conclusions about its effectiveness [52]. The review has not been updated since.

Since 2003, several reviews have examined related areas of MS rehabilitation—including telerehabilitation, multidisciplinary programmes, and broader domains of OT practice—but none has offered a comprehensive and up-to-date evaluation of controlled trials specifically targeting OT interventions for MS. This highlights the need for a review that systematically assesses the current evidence base for occupational therapy in this population. Our review is not an update of the Cochrane review by Steultjens and colleagues [52]; it should be considered as a new review given the extent to which we have changed the methods to reflect current standards.

The evidence base for OT has grown significantly in recent decades, as can be seen from a 2021 scoping review in the

area [53]. At the same time, OT practice itself has evolved, shifting from the predominantly biomedical approach to a more client-centred, holistic and context-sensitive approach [53, 54, 55, 56]. Modern OT increasingly addresses not only the needs of the person with MS but also those of caregivers and the wider physical and social environment, with greater emphasis on supporting participation through home- and community-based interventions [57, 58, 59].

There is therefore a clear need for a current synthesis of the OT evidence to support therapists who want to ensure that their practice is evidence-based and for people with MS to have an impartial resource to assist them in decision-making. Researchers may benefit from this review as it has identified gaps in knowledge and provides suggestions for future research in OT for people with MS. The results of this review are also relevant to other stakeholders, including MS caregivers, policymakers, guideline developers and health insurance organisations.

OBJECTIVES

Main objective

To assess the benefits and harms of occupational therapy interventions for improving daily functioning, participation and quality of life in people with multiple sclerosis.

Secondary objective

To assess whether the effects of occupational therapy interventions differ according to the format of intervention delivery (individual versus group) and location of service delivery (outpatient, inpatient or home-based therapy).

METHODS

Differences between protocol and review

The review deviated from the protocol in the following ways.

- Eligibility criteria: in terms of types of interventions, we stated in the protocol that "When it is not clear if it is an occupational therapy (OT) intervention delivered by an occupational therapist, we will include studies on interventions that aim to support people with MS to perform or cope with their daily roles and tasks, according to the definition of occupational therapy (World Federation of Occupational Therapists 2012)" [60, page 4] ([Description of the intervention and how it might work](#)). Following the planned piloting phase of the eligibility criteria of 50 records, we narrowed the eligibility criteria of the types of interventions and focused on interventions explicitly involving an occupational therapist, with the role of the occupational therapists clearly specified in the study. This decision also aligns with the research priority of the James Lind Alliance, to "explore the unique contribution that occupational therapy makes to people who access services and the impact it has on their everyday lives" [51].
- Data extraction: we did not conduct the extraction of study characteristics data as initially planned. Instead of two review authors independently extracting data, one review author (CM) performed data extraction, and two others (JF, DK) verified the extracted data. However, as planned, two review authors (AB, JF, IE or DK) independently extracted data of outcomes of each study, supervised by the methodology expert (GB).

- Search strategy: we slightly changed the mesh term #7 ("neuromyelitis optica":ab,ti) in the search string for Embase, as it did not recognise the letter 's' in two terms. We removed the apostrophe (') in the Embase search string #9 ('devic disease':ab,ti OR 'Devic Syndrome':ab,ti OR 'Devic's Syndrome':ab,ti OR 'Devices Syndrome':ab,ti OR 'Devic Disease':ab,ti OR 'Devic's Disease':ab,ti OR 'Devices Disease':ab,ti), as this term was not recognised ([Supplementary material 1](#)). Additionally, we found an error in the search string for CENTRAL, where term #44 was duplicated: #44 dexter*:ti,ab,kw and #44 "joint protection":ti,ab,kw. After correction, the search string for CENTRAL consisted of 50 terms instead of 49.
- Search sources: we omitted some of the less important sources we had planned to search, and we added some others that we thought might be fruitful.
- Deduplication of search records: in the protocol, we specified that we would follow Bramer's method using "EndNote software" without naming a specific version [61]; however, the

published Bramer workflow was demonstrated using EndNote 2015. For the current review, we used EndNote 20 [62], which supports the same deduplication functions. Manual checking for duplicates was required, as a few duplicate records persisted even after EndNote and Covidence screening [62]. These were removed manually during title/abstract screening.

Criteria for considering studies for this review

Types of studies

We included randomised and non-randomised controlled studies addressing occupational therapy for people with MS, including the first phase of cross-over studies (to avoid carry-over effects from the initial intervention phase). Non-randomised studies were eligible for inclusion because of the anticipated dearth of published randomised controlled trials (RCTs) [53], but we also included non-randomised studies of interventions to help ensure that we conducted and reported an equity-oriented systematic review, as described in the *Cochrane Handbook for Systematic Reviews of Interventions* and in the logic model pictured in [Figure 1](#) [63, 64].

Figure 1. Logic model – potential impact of health (in)equities

Figure 1 Logic Model – potential impact of health (in)equities

What is known?	Factors- the Person (Christiansen 2015)	Factors- the Environment (Christiansen 2015)	Factors- the Occupation (Christiansen 2015)	Potential Impact on Review	How will this be addressed?
• Occupational therapy globally-estimated to be 527,997 occupational therapists across 88 countries. Range 0.0002-16 per 100,000 head of population (WFOT 2018). • MS treatment and diagnosis- 7 out of 10 (of 82% countries globally) countries face barriers in access to disease modifying treatments (DMT); particularly high barriers in low income countries. 100% of low income countries do not use high efficacy licenced DMTs. 83% of countries report barriers to early diagnosis (MSIF 2021). • MS rehabilitation globally-high variability across countries. Less availability in lower income countries (MSIF 2021).	• Psychological • Accessibility and utilisation of health-care services (Dover 2019). • Cognition • Access to cognitive rehabilitation varies greatly across countries (MSIF 2021). • Neurobehavioural • Impact of pre-existing health status. • Motor • Those with greater disability less likely to access services (Roddam 2019). • Physiological • Males less likely to access services (Roddam 2019). • Older people less likely to see neurologist (Roddam 2019). • Meaning/Sense-Making/Spiritual • Health beliefs affect health-related behaviours (Dover 2019).	• Education and Policy • Those with fewer years of education less likely to access services (Roddam 2019). • Impact of social and public policies on access to health, education, labour market (Solar 2010). • Physical and Natural • Difficulty accessing services for people in non-urban areas (Roddam 2019). • Culture • Societal values (Solar 2010). • Social Determinants • Socioeconomic and political context (Solar 2010). • Education, occupation, income, gender, ethnicity (Solar 2010). • Social Support and Social Capital • Practical, information, and emotional support. • Assistive Technology	• Often categorised as activities, tasks, and roles that we need to do, want to do or have to do (Christiansen 2015). • Criticisms of a dominance of theories and models that come from English-speaking minority countries (Hammell 2019). This may be applicable to the definition of occupational therapy that guides this review, and focus of intervention. • Many factors determine participation in occupations, e.g.: • Components of health (in)equality determine access to occupational opportunities (Dover 2019). • Those who are unemployed have increased barriers to attending medical appointments (Roddam 2019).	• Health policy context could impact on the acceptability, appropriateness, availability, and effectiveness of the intervention (occupational therapy) provided (Dover 2019). • Utilisation of health-promoting resources across settings (Dover 2019). • In low/middle income countries the high unmet need for rehabilitation indicates that research found is likely to be from high income countries (MSIF 2021). • Evidence of inequalities in access to health services in MS (Roddam 2019). Though not clear for occupational therapy and MS, research, this needs to be considered. • Recommendations made in the context of known inequities.	• Including non-randomised studies in review, as recommended (Tugwell 2010). • Evaluate and report baseline imbalance across PROGRESS-Plus factors (O'Neill 2014). • Contexts listed in this model will be considered in review findings. • Health inequity components will be included in summary of findings table-separate row for differences (Welch 2022), if found. • Will interpret findings related to health equity in the discussion, including impact on intervention and review outcomes. • Recommendation(s) will consider health inequity in research and clinical practice. • Complete checklist (Ueffing 2010).

To be included, non-randomised studies had to be prospective with a comparison group, such as controlled clinical trials or cohort studies (including case-cohort or nested case-control studies). We excluded retrospective studies with a comparison group, such as retrospective cohort studies or case-control studies. We also excluded studies without a comparison group, such as case series, as well as records that did not report any study results (e.g. protocols, conference abstracts without data or descriptive reports without outcome data).

Types of participants

We included studies of people over the age of 18 years who had a diagnosis of MS (as described in each study). We included all types of MS and all disease durations.

We excluded studies if they involved people with mixed diagnoses unless results were reported separately for the subgroup of people with MS, or if the proportion of people with MS was at least 75% of the total population.

Types of interventions

We included studies in which authors reported on the effects of 'occupational therapy', that is an occupational therapy intervention or an intervention delivered by an occupational therapist. When this was not clear from the reporting of the study, we included interventions explicitly involving an occupational therapist (with the role of the occupational therapist clearly specified in the study) who was aiming to support people with MS to perform or cope with their daily roles and tasks, according to the definition of occupational therapy in the [Background](#) section [32]. We included studies that took place in hospital, rehabilitation and community settings.

We placed no restrictions regarding intensity of the intervention (frequency (sessions per week), length of session (minutes) and intervention duration (days/weeks)) or mode of delivery.

We allowed for the inclusion of co-interventions such as disease-modifying treatments, regular physical therapy or others, if they were provided to both the intervention and comparison groups. We excluded studies addressing multidisciplinary programmes unless they specifically evaluated the unique contribution of occupational therapy in the multidisciplinary approach.

We included studies that compared occupational therapy with an active control intervention, standard care/practice or no intervention, which we defined as follows.

- Active control intervention: the control group received a specific control or placebo intervention, which went beyond usual care and was designed to actively engage participants, allowing for a meaningful comparison with the experimental intervention.
- Standard care/practice: the control group received usual care, defined as the standard clinical practice routinely provided in the given setting, without additional structured interventions.
- No intervention: participants in the control group received no intervention or were placed on a waiting list and received the intervention after the study period.

We also included studies directly comparing occupational therapy with another intervention such as a behavioural, physical or pharmacological intervention.

Outcome measures

Outcomes in occupational therapy are person-driven and are typically assessed by examining satisfaction with activity performance, participation, and improvements in engagement in daily activities [65]. Given this, we included studies that reported on any of the outcomes listed below, and we allowed variants of instruments and outcome measures. The instruments listed below were agreed at the protocol stage.

We included up to three predetermined time points for measuring each outcome, including post-intervention, and up to two follow-up assessments after the intervention (medium term: up to six months; long term: longer than six months), depending on what was reported in the studies.

We used the following hierarchy to select one outcome per domain for the analysis: 1) an outcome instrument listed in [Critical outcomes](#) and [Important outcomes](#); 2) an outcome listed in the core outcome set for people with MS [66]; 3) the study's primary

outcome; and 4) the outcome most closely related to the outcome domain within the scope of the specific intervention.

To improve interpretability, for all outcomes, we reversed the direction of outcomes used in analysis where necessary. So, for all outcomes, higher scores suggest better performance or less impact from MS.

Critical outcomes

Daily functioning

The Canadian Occupational Performance Measure (COPM) evaluates outcomes in the areas of self-care, productivity and leisure that are identified by the person completing the questionnaire [67, 68]. First, the most important problems are identified, then individuals score their ability as they see it and their satisfaction with that performance. Two final scores are obtained: one for performance and one for satisfaction with performance, both on a scale from 1 to 10, with higher scores representing better performance or higher satisfaction. Suggested thresholds for clinically important change (in an outpatient population) range between 0.9 and 1.9 [69]. Several studies have used the COPM to describe occupational performance in people with MS [70, 71, 72].

Quality of life

Health-related quality of life (HR-QoL) refers to how a person's physical condition and psychological well-being affect their ability to function and participate in daily life. It includes aspects such as mobility, pain, emotional state, cognitive function and social engagement and is often expressed as physical HR-QoL and mental HR-QoL [73].

The Multiple Sclerosis Impact Scale (MSIS-29) evaluates the physical (20 items) and psychological (9 items) impact of MS from the perspective of the person with MS [74]. All items use the Likert scale scoring format (range 1 to 5). Sum scores for the two subscales are converted to a 0-to-100 scale, with higher scores representing more impact. The suggested threshold for a clinically significant change in physical impact is 7.5 points [75]. MSIS-29 is one of the core outcome measures for quality of life in exercise studies in people with MS [76]. We used the subscales of the MFIS-29 to assess physical and mental HR-QoL.

Adverse effects

To assess adverse effects, we took an exploratory approach [77]. We reported the type and frequency of any reported adverse events occurring during the trial or follow-up (across intervention groups). These included but were not limited to: falls, need for medical intervention, need for hospitalisation, MS relapse or exacerbation, mortality and morbidity. When no adverse events were reported in the study, we documented this.

Important outcomes

Participation

The Impact on Participation and Autonomy (IPA) Questionnaire evaluates self-reported participation (31 items) and perceived problem (8 items) for eight subdomains [78, 79]. Perceived participation items are scored from 1 = excellent to 5 = very poor. Perceived problem items are graded from 0 (no problem) to 2 (severe problem). The sum score for each domain can be calculated, with a higher score representing a greater perceived

restriction in participation or problem. IPA has been used to predict participation and autonomy in 194 people with MS [80].

Resilience

The Connor-Davidson Resilience Scale (CD-RS) measures how well a person can cope with stress. It consists of 25 items, each scored from 0 to 4. Sum scores range from 0 to 100, with higher scores reflecting greater resilience [81]. Several studies have used the CD-RS to explore resilience in people with MS [82, 83, 84].

Self-efficacy

The Multiple Sclerosis Self-Efficacy Scale (MSSE) measures self-efficacy specifically in individuals with MS [85, 86, 87, 88]. The original version contained 24 items, but the revised version retained 18 items scored from 10 to 100 (anchored by 10 = very uncertain, 50 = moderately certain and 100 = very certain). There are two subdomains: function and control (9 items each). MSSE is used to study self-efficacy in people with MS.

Self-management

The Multiple Sclerosis Self-Management Scale (MSSM) addresses self-management knowledge and behaviour amongst individuals with MS [89, 90, 91, 92]. The scale has 24 items and five factors (healthcare provider relationship/communication, treatment adherence/barriers, social/family support, MS knowledge and information, and health maintenance behaviour). Items are scored on a Likert-type scale ranging from 1 (completely disagree) to 5 (agree completely). Higher scores mean better self-management.

Mood

The Hospital Anxiety and Depression Scale (HADS) measures feelings of anxiety and depression without reference to the source of the complaints [93, 94]. It consists of 14 items: 7 each for the two subscales of depression and anxiety. Each item is scored on a 4-point scale (0 to 3). Total scores for the subscales range from 0 to 21, with higher scores representing more depression and anxiety. Scores between 0 and 7 for each subscale are within normal ranges. HADS is frequently used as a screening tool for depression in MS and to describe relationships with other domains, like participation [95, 96].

Impact on caregivers

The Caregiver Strain Index (CSI) examines subjective and objective elements of caregiver strain and includes 13 items to be scored 0 (no), 1 (sometimes) or 2 (on a regular basis) [97]. Thornton and Travis adapted the CSI for relevance to long-term caregivers, which also improved its psychometric properties [98]. Several studies have used CSI to describe the perspective of caregivers of people living with MS [99, 100].

Search methods for identification of studies

We adhered to the PRISMA 2020 statement guidance [101]. Two review authors (AB, CM) conducted all searches, following and applying the eligibility criteria established for the review. We conducted an extensive computerised search to address the threats of publication bias, language bias and to identify as much relevant literature as possible. The Participant, Intervention, Comparison and Outcomes (PICO) for the search were defined by the eligibility criteria. The planned search drew on appropriate subject headings and text words based on the eligibility criteria.

The final search strategy consisted of three key concepts: multiple sclerosis (P), occupational therapy (I), and study design and included all controlled trials. We did not include a comparator in the search terms, to reduce the chance of inadvertently excluding relevant records. To increase precision and the generalisability of the review results, we did not restrict the search by date or language.

We maintained a detailed record of the search process (see [Supplementary material 1](#)). This record includes the sources searched, when the search was conducted, by whom and the search terms used [102]. For grey literature (search strategy detailed below), we recorded the URLs of all websites and kept a record of sources accessed. We also recorded reference lists searched and investigators contacted as part of the search process. We followed guidance from the PRISMA-Search (PRISMA-S) Extension in the reporting of the search [103].

We re-ran the search of bibliographic databases on 17 November 2024. We did not update the searches of the trial registries because registry records were not expected to yield additional completed trials with available results relevant to our inclusion criteria. An additional registry search was therefore unlikely to influence the findings. We checked whether any protocols identified in the initial search had since been published in the updated database search. We incorporated new findings prior to the publication of our review.

Electronic searches

Two review authors (AB, CM) completed a systematic search of the following electronic databases on 17 November 2024.

- Cochrane Central Register of Controlled Trials (CENTRAL) via Wiley Online Library (from inception 1996 to November 17th 2024)
- MEDLINE via PubMed NLM (1946 to 17 November 2024)
- Embase via Elsevier (1974 to 17 November 2024)
- CINAHL via EBSCOhost (1982 to 17 November 2024)
- PsycInfo via EBSCOhost (1800 to 17 November 2024)
- Web of Science via Clarivate (1975 to 17 November 2024)

The search strategy, developed in consultation with an Information Specialist, can be found in [Supplementary material 1](#) [104], along with a record of the databases searched, the interface used and the date ranges of the search.

Searching other resources

Two researchers (AB, CM) screened the reference lists of included studies. We screened reference lists of similar reviews in the area [52, 53, 54, 55, 105, 106]. We examined any retraction statements and errata of included studies by querying the original DOIs (digital object identifiers) on the relevant publisher websites and did not find any retractions or errata.

We searched the following clinical trial registers on 24 July 2023: ISRCTN registry (<https://www.isrctn.com/>), ClinicalTrials.gov (<https://www.clinicaltrials.gov/>) and the World Health Organization International Clinical Trials Registry Platform (<https://www.who.int/clinical-trials-registry-platform>). Details of our search terms are provided in [Supplementary material 1](#).

We searched the grey literature, specifically research theses, reports and conference abstracts. Specifically, we searched the Institute for Scientific and Technical Information database (on 30 November 2023), which includes all records in greynet.org and all records previously hosted on the System for Information on Grey Literature database. In addition, we searched the Healthcare Management Information Consortium database (on 21 December 2023), the National Technical Information Service (on 30 November 2023) and PsycEXTRA (on 4 December 2024). We also searched research newsletters of multiple sclerosis societies (on 30 July 2023). Details of our search terms are provided in [Supplementary material 1](#).

Data collection and analysis

We used EndNote reference manager software to manage the retrieved records [107] and Covidence software to support and manage the review process [62]. We used Review Manager (RevMan) to conduct analyses [108].

Selection of studies

One review author (CM) first merged the studies from all sources into EndNote software and removed duplicates following the steps described by Bramer and colleagues [61]. The deduplicated records were then imported into Covidence software, where additional duplicates were identified using Covidence's automated deduplication function. During screening in Covidence, remaining duplicates that had not been detected by either software were identified manually and marked as duplicates.

Two review authors (AB, CM) then piloted the eligibility criteria, by independently screening 50 records. Based on the results of the pilot, we made the following clarifications.

The pilot study revealed insufficient clarity regarding what was considered an occupational therapy intervention for review authors with different backgrounds. To address this uncertainty, we clarified the eligibility criteria such that included studies must have explicitly mentioned the involvement of an OT. Following this clarification, we discussed and re-screened the 50 studies.

Another point of discussion within the review team was whether to include multidisciplinary interventions. We adhered to the review's protocol by including only those multidisciplinary programmes where the OT's role was clearly defined and results were distinctly reported. It is important to acknowledge that studies in which the occupational therapist's role was not clearly delineated within a multidisciplinary intervention may have been excluded.

Two review authors (AB, CM) independently screened the abstract and title using the eligibility criteria. The two review authors met online to discuss disagreements. If no consensus was reached, we consulted a third review author (LD) to resolve any disputes.

Following this, we retrieved the full texts of all records deemed potentially relevant. Three review authors (AB, CM, LD) screened these using the same eligibility criteria. When multiple reports from the same study were available, we used information from all reports that contributed relevant outcome data. To avoid double-counting, we identified related reports through trial registration numbers or by contacting authors, and we included only the results based on the full study sample.

If it was unclear if a study was conducted by an occupational therapist, if the intervention was occupational therapy or what the outcomes were, we corresponded with the trial investigators, where possible. The final decision on inclusion was decided by a minimum of two review authors working independently (AB, CM, LD), and we managed disputes through an online meeting with all review authors to resolve any uncertainties or misunderstandings. We kept a record of ongoing trials that met the inclusion criteria. These records were revisited prior to publication of the review to check whether their publication status had changed.

The review authors involved in study selection had no conflict of interest (i.e. they had not worked on or published any study that could meet the review eligibility criteria) [60].

Data extraction and management

Each review author involved in data extraction used the data extraction form developed specifically for this review. Two review authors (AB, CM) piloted the data extraction form with six studies to ensure uniformity of extraction between review authors. The authors discussed experiences in an online meeting and made changes accordingly. For each included study, two review authors extracted the outcome data independently (AB, DK, JF, IE). Review authors who were authors of an included study (DK, SH, IE) were not involved in the extraction of their own study data. CM extracted study characteristics, which were checked by another member of the review team (JF, DK or IE). We discussed any disagreements in data extraction in online meetings with a third member of the review team (LD). We contacted study authors if outcome data were missing.

The data extraction tool included the following data.

- Source: study ID; report ID; review author ID; citation and contact details; funding details
- Eligibility: confirmation of eligibility or reason for exclusion
- Study design: RCT/controlled clinical trial (CCT); total study duration
- Participants: age; sex; occupation; year of diagnosis; MS phenotype; degree of disability; total number randomised; diagnostic criteria; comorbidity; race/ethnicity; geographical area; inclusion and exclusion criteria
- Potential effect modifiers: disease duration; baseline level of disability; age and comorbidity
- Intervention: description of the intervention (including intervention details); description of the comparison interventions; intervention setting; geographical location; professional delivering the intervention; caregiver involvement; number of intervention groups; duration and frequency of the intervention and follow-up; blinding; integrity/fidelity of the intervention. As can be seen in the data extraction tool, we used the Template for Intervention Description and Replication (TIDieR) for the description of study interventions [109].
- Outcomes (collected and reported): critical outcomes (daily functioning, quality of life, adverse effects), and important outcomes (participation, self-efficacy, self-management, mood, resilience, impact on caregivers); specific outcome measure or person-reported outcome used. This includes outcome definitions, units of measurement, and upper and lower limits, where applicable.

- Results: number of participants allocated to each group; numbers included in the analysis; estimate of effect for each outcome with confidence interval; subgroup analyses

In addition to the data extraction form, we created a 'Characteristics of included studies' table ([Supplementary material 2](#)), and a 'Characteristics of excluded studies' table for all studies that we screened in full text but did not include in the review. We recorded the reasons for exclusion (see [Supplementary material 3](#)).

Risk of bias assessment in included studies

Two review authors (AB, LD) independently performed risk of bias assessments using the Cochrane RoB 2 tool [110]. We compiled the ratings in the Excel tool to implement RoB 2. Importantly, this tool assesses the risk of bias per outcome and not per study [111]. We assessed results for all outcomes separately ([Outcome measures](#)), for short-term (post-intervention), medium-term and long-term follow-up. For cross-over trials, we included the first period when both groups started at the same time. As outlined in the tool, and included in the risk of bias table, we (AB, LD) assessed five domains for risk of bias: 1) randomisation process, 2) deviations from the intended intervention, 3) missing outcome data, 4) measurement of the outcome and 5) selection of the reported result.

Review authors independently judged each outcome and the overall risk of bias as 'low risk', 'some concerns', or 'high risk' [110]. Overall results had at least the same risk of bias (if not more severe) than any of the individual domains assessed [111]. We focused on the effect of assignment to the interventions at baseline, regardless of whether the interventions were received as intended (i.e. the 'intention-to-treat effect') [111]. During the consensus discussion, we considered the use of self-reported outcome measures for domain 4 (bias in measurement of the outcome). Based on careful deliberation and the nature of the interventions, we decided to deviate from the algorithm and not assign a high risk of bias for domain 4 to any of the included studies reporting on any of the outcomes other than daily functioning. The rationale for this decision is that the self-reported measures used are psychometrically sound and commonly used in occupational therapy and other health domains for outcomes such as quality of life, participation, self-efficacy, self-management, resilience and mood. These outcomes are based on a person's own experience, which can only be measured by self-report. Worldwide, the regular use of patient-reported outcome measures (PROMS) for evaluating health care is growing in rehabilitation trials, as well as in pharmacological trials [112, 113]. However, for daily functioning and particularly when fatigue measurements were used, we judged a high risk of bias to be appropriate due to the potential for symptom focusing [114].

For the results of included non-randomised studies, we used the Risk of Bias In Non-randomised Studies of Interventions (ROBINS-I) tool to determine the risk of bias for each outcome [115]. Again, we evaluated results of all important outcomes separately, for short- (post-intervention), medium- and long-term follow-up. This tool consists of seven domains of concern: four of the RoB 2 domains (i.e. 2, 3, 4 and 5), as well as confounding, selection bias and measurement classification of interventions. For each outcome, the studies are categorised as 'low risk', 'moderate risk', 'serious risk' or 'critical risk' of bias. In line with our protocol, the main confounders that we considered when assessing risk of bias due to confounding in non-randomised studies included disease

duration, baseline level of disability, age, comorbidity, baseline functional performance and social support [116, 117]. We selected these factors based on their potential to influence the outcomes of interest, and we extracted them from included studies when available. This information guided our use of the ROBINS-I tool and helped ensure a consistent and transparent approach to evaluating confounding across different outcome types.

We analysed the effects of the interventions in the groups to which they were originally randomised, regardless of their level of adherence to the intervention protocol or whether they received the intervention as intended ('intention-to-treat principle') [111]. Co-interventions were allowed (e.g. disease-modifying therapy, physical therapy) if they were provided to both the intervention and comparison groups.

Any disagreements were discussed and resolved by consensus. The review team kept a record of the risk of bias table with all decisions made in relation to the risk of bias in each domain, along with a rationale for each decision made.

Measures of treatment effect

The review team made the decision that a meta-analysis was appropriate, based on data extraction and review of the availability of usable data including clinical characteristics of the studies and assessment of risk of bias.

For continuous outcomes, such as daily functioning, quality of life or participation, we used the standardised mean difference (SMD) where outcomes were not all measured using the same scale. The SMD allowed for comparison across studies and provided an expression of the size of the effect of the intervention, relative to study variability, with 0.2 representing a small effect, 0.5 a moderate effect, and 0.8 a large effect [118, 119]. When studies used the same scales, we used the mean difference (MD). We calculated 95% confidence intervals (CI) as a measure of precision.

For dichotomous outcomes such as the adverse effects, we used the risk ratio (RR).

Unit of analysis issues

Where studies compared occupational therapy with more than one comparison group, we selected the group that was most similar to usual care as our main control group.

As cross-over trials are not suitable for the intervention of our review due to anticipated long-lasting effects, we only considered the first period of such trials. Subsequently, we treated cross-over trials as standard RCTs.

Dealing with missing data

Where possible, we assessed the extent of data that were missing from included studies. Where studies used imputation for missing data, the review team discussed this and decided whether the assumptions made were plausible or if they had the potential to bias the study results. We used existing guidance in this area [120].

Where standard deviations (SD) were missing, we calculated the SD based on the confidence interval or interquartile range [121, 122].

Reporting bias assessment

In accordance with our protocol [60], as there were fewer than 10 studies in the meta-analyses that we conducted, we did not evaluate the possibility of non-reporting bias by means of contour-enhanced funnel plots [60123].

Synthesis methods

We conducted our analysis using RevMan [108].

Given the broad nature of the intervention (occupational therapy) in this review, and the resultant clinical heterogeneity (e.g. differences in intervention, intensity, duration) and methodological heterogeneity (e.g. differences in outcomes, trial design), we used a random-effects model in the meta-analysis. We took a two-step approach for each study, whereby we calculated a uniform measure of effect (SMD, RR or MD with 95% CI). We used the Wald method to calculate summary effect confidence intervals [124]. We then pooled the results of individual studies. For continuous results on the same outcome but reported using different scales, we pooled studies using the SMD.

We selected outcomes and effect measures for the analysis (see [Criteria for considering studies for this review](#)). For the meta-analysis, we applied the inverse method employing random-effects meta-analysis, using the DerSimonian and Laird method [124]. Using this approach, each study was weighted as the inverse of their effect estimate. As a result, larger studies, which have smaller standard errors, are assigned larger weights compared to smaller studies, which have larger standard errors. We used forest plots to present the results of the meta-analyses.

In a separate analysis, we assessed non-randomised studies in detail with respect to the PICO elements before data synthesis, as suggested in Section 24.6.2 of the *Cochrane Handbook* [63]. We only pooled the results of non-RCTs if the overall risk of bias was low to moderate and adjusted data were presented for at least baseline level of disability, disease duration and comorbidity [116, 117].

All data that were not included in meta-analysis are reported in narrative form and summarised in tabular format, according to Synthesis Without Meta-analysis (SWiM) guidelines [125].

We interpreted results as described in chapter 15 of the *Cochrane Handbook* [126]: we based our conclusions on the effect size of the pooled value (determining whether there was an effect using predefined thresholds (SMD 0.2 = small effect, 0.5 = moderate effect and 0.8 = large effect [127])) and the certainty of evidence.

Investigation of heterogeneity and subgroup analysis

We initially assessed heterogeneity using visual inspection of a forest plot and the χ^2 test included in the plot. We then quantified heterogeneity using the I^2 estimate. We provided a narrative description of any heterogeneity across the included studies, along with potential explanations for its occurrence.

In line with our secondary objective, we planned two subgroup analyses that are of clinical importance to the review question: 1) differences in the format of intervention delivery (individual versus group format) and 2) differences in the effects of the intervention across the locations of service delivery (inpatient, outpatient or day treatment, and home-based settings). Both analyses are of

importance to clinicians (see [Description of the intervention and how it might work](#)).

We did not perform our planned subgroup analyses due to the limited number of studies for each outcome.

Equity-related assessment

We developed a logic model at the protocol stage [60] to understand the potential impact of health equity on the intervention and listed health equity components of importance to the review [128, 129, 130, 131]. See [Figure 1](#). We considered health (in)equity in the review by:

- including non-randomised studies [64, 132];
- extracting data relating to participant factors that may result in inequitable access to interventions using the PROGRESS-Plus framework [133]. We extracted data related to place of residence, race/ethnicity, language, education, sex/gender, religion, socioeconomic status, social capital, and data related to personal characteristics potentially associated with discrimination (e.g. age or disability), where reported;
- evaluating baseline imbalance across PROGRESS-Plus factors [133];
- including health inequity components in a summary of findings table - a separate row is reported for differences [132];
- interpreting findings related to health equity in the discussion, including the impact on intervention and review outcomes [132];
- considering health inequities in research and clinical practice in the recommendations provided; and
- using the Equity Checklist for Systematic Review Authors [134] ([Supplementary material 8](#)).

To check for potential equity concerns in the included studies, we examined domain 1.3 of the RoB 2.0 tool, which evaluates whether baseline differences between intervention groups suggest a problem with the randomisation process. We specifically checked whether studies conducted subgroup analyses or accounted for baseline differences, and how these differences might have influenced the outcomes. This is important from an equity perspective, as systematic differences at baseline can lead to biased results and affect certain subgroups.

Sensitivity analysis

We conducted a sensitivity analysis, using RevMan [108], excluding studies with high risk of bias to explore the influence of risk of bias on the overall conclusions made and precision of the results. Specifically, we applied sensitivity analysis through the exclusion of studies that are considered to have an overall high risk of bias (as per Cochrane RoB 2 tool). We discuss these results in the [Synthesis of results](#).

Certainty of the evidence assessment

Certainty of evidence

Three review authors (AB, LD, GB) rated the certainty of evidence for each critical outcome based on the five GRADE domains [135, 136, 137]. We categorised the evidence for each outcome as high, moderate, low or very low, and we documented the justification for each rating using GRADEpro GDT [138]. We followed the guidance provided in Chapter 14 of the *Cochrane Handbook* [139] - results

of the RoB 2/ROBINS-I assessments fed directly into the GRADE assessment domain risk of bias: 'low' risk of bias indicated 'no limitation'; 'some concerns' indicated either 'no limitation' or 'serious limitation'; and 'high' risk of bias indicated either 'serious limitation' or 'very serious limitation'. 'Critical' risk of bias on ROBINS-I indicated extremely serious limitations in GRADE.

Summary of findings

We produced a summary of the findings table using GRADEpro GDT software [138]. As suggested by Schünemann and colleagues [139], we included the key information from the review for each outcome: the number of studies, the number of participants included in the analysis, and the absolute and relative magnitude of the effect measured. All comparators (active controls, treatment as usual or no intervention) are described in the table along with the critical outcomes (daily functioning, physical and mental health-related quality of life, and adverse effects) and the important outcome with most study results (participation). We presented a GRADE assessment of the overall certainty of the body of evidence and included explanations, using the guidance provided by Santesso and colleagues [140].

Consumer involvement

One of the co-authors (MK) represented people living with multiple sclerosis, based on her own experiences as well as discussions with

peers resulting from her active involvement in the national patient organisation. MK was involved throughout the review process: protocol development, discussing review conduct and results, and writing and reviewing the review manuscript.

RESULTS

Description of studies

[Supplementary material 2](#) contains the characteristics of the included studies. [Supplementary material 3](#) lists excluded records, each with the reason for their exclusion.

Results of the search

Through electronic searches, we identified 12,570 records. In addition, we found 278 records in grey literature. In the updated search in November 2024, we identified one publication of a study that was on the list of ongoing trials in the first search [170]. After de-duplication, we discarded irrelevant records. Of the remaining records, we excluded 194 records and presented the reasons for these exclusions in the review, and we identified 3 ongoing studies. Finally, we included 20 studies (24 records) in this review. See the PRISMA flow chart for details and reasons for exclusion ([Figure 2](#)); see [Supplementary material 4](#) for details of ongoing studies.

Figure 2. Results of study selection based on records identified by search strategy

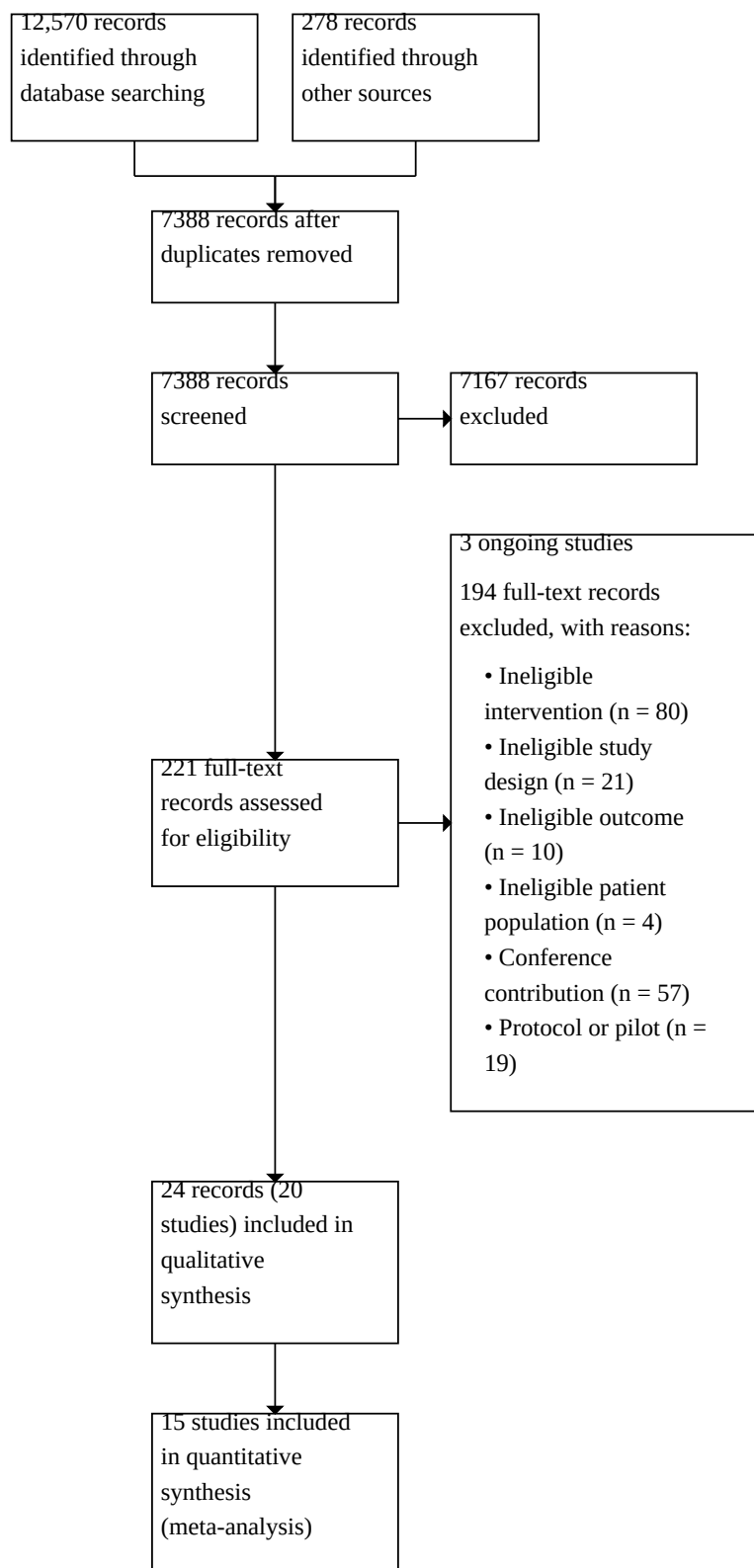


Figure 2. (Continued)

(meta-analysis)

We contacted eight study authors via email for missing or inconsistent data (or both) [141, 142, 143, 144, 145, 146, 147, 148]. Five authors provided the necessary data in time for our analysis [143, 145, 146, 147, 148].

Included studies

We included 20 studies: 17 RCTs and three controlled clinical trials, which included 1628 participants with MS (Supplementary material 2). The key characteristics of studies and the synthesis methods used in this review are summarised in Table 1.

Characteristics of samples

The studies in this review included adults with multiple sclerosis who were between 18 and 70 years old. Most of them had low to moderate levels of disability. The ratio of women to men in the studies was 3:1. Nineteen of the 20 studies were performed in high-income countries; one study was conducted in Iran. More details of sample characteristics are provided in Supplementary material 2.

Types of interventions and comparisons

Interventions

Ten studies evaluated a fatigue management programme, which was delivered in group or individual format, using face-to-face or remote delivery, or using a combination of both formats and types of service delivery [141, 143, 144, 145, 147, 149, 150, 151, 152, 153].

Nine studies evaluated occupational therapy interventions aimed at enabling daily functioning, including training of skills [154, 155, 156, 157], enabling self-management [142, 155, 158, 159] or empowering people to deal with their cognitive difficulties in daily activities [146].

One study evaluated a socialisation programme (Community Reintegration for Socially Isolated Patients (CRISP)) to improve social participation in socially isolated people living with MS [148].

Context

Six of the interventions evaluated in the studies were delivered in group format [143, 144, 148, 150, 152, 153], 11 were delivered to individuals [141, 142, 147, 149, 154, 155, 156, 157, 158, 159, 160] and three used a combined approach [145, 146, 151]. With the different comparison groups (i.e. active control, usual care and no intervention), we did not reach a sufficient number of studies for subgroup analysis for any outcome. Therefore, we did not perform the planned subgroup analyses.

The OT interventions were delivered in different settings: eight were in the community or were home-based [143, 146, 150, 152, 153, 157, 158, 159], four were in outpatient settings [142, 149, 154], two in an inpatient setting [145, 147] and four used a combination of outpatient and home-based therapy [141, 148, 151, 160]. Three studies did not report the setting [144, 155, 156].

Comparisons

Comparisons were an active control intervention (information provision, "placebo" or alternative programme) [145, 147, 149, 151, 155, 158, 160], usual care [141, 142, 144, 146, 148, 152, 153, 154, 157] or no intervention (waiting list) [143, 150, 156, 159].

Outcomes

In Table 1, we summarised the outcome measures of all included studies. All studies used a variety of self-reporting instruments assessing at least one of our critical or important outcomes (Outcome measures). Two studies used an observer-based measurement to assess daily functioning [155, 156]. All studies collected data within one week following the intervention. Seven studies also included a medium-term follow-up [142, 145, 146, 149, 151, 153, 157], one of which used this as the primary time point [146]. Two studies provided long-term follow-up of up to one year [149, 161].

Twelve studies evaluated outcomes other than daily functioning and quality of life [143, 144, 145, 146, 148, 149, 150, 151, 153, 155, 158, 159].

None of the studies included in this review systematically reported adverse events or the impact on caregivers. Eight studies explicitly stated that there were no adverse events related to the intervention or the study. In some studies, reasons for dropout were reported (e.g. hospitalisation, MS relapse, emergency surgery, lack of time) but these were not categorised as adverse events. Moreover, the studies did not report how these were assessed or monitored using a systematic method. No study reported the frequency of adverse events. Only one study reported using 'questions' for capturing adverse events, without further explanation [151]. Therefore, there are very limited data on which to base any insight into our planned outcome of 'type and frequency of any reported adverse events occurring during the trial or follow-up', regardless of their relation to the intervention (see Critical outcomes).

Excluded studies

Based on the full-text screening, we excluded 197 studies for various reasons (Supplementary material 3; Figure 2). The most frequent reason was that the intervention did not meet the criteria for occupational therapy as described in our Methods (see Types of interventions). We excluded other studies because they reported only outcomes outside the scope of this review (e.g. fatigue severity, mobility, cognitive function or manual dexterity function instead of our targeted outcomes of daily functioning, QoL or participation), or because they used inappropriate study designs (e.g. observational studies or case series). We excluded a number of studies available only as conference abstracts or posters. Additional exclusions were due to ineligible patient populations or lack of published results.

Risk of bias in included studies

We used Cochrane RoB 2 for RCTs and ROBINS-I for non-randomised studies for each available outcome and time point [110, 115]. In [Supplementary material 5](#), we provide an overview of the risk of bias results. Forest plots showing the risk of bias for each study outcome are provided in [Supplementary material 6](#). Below, we summarise the risk of bias for the outcomes reported in the summary of findings tables: daily functioning, quality of life, adverse events and participation. In addition, we provide a summary of the outcomes not included in the summary of findings tables: resilience, self-efficacy, self-management, mood and impact on caregivers.

Daily functioning

Eleven RCTs included in the meta-analysis reported on the outcome of daily functioning [141, 143, 145, 146, 147, 149, 151, 155, 158, 159, 160].

We had some concerns about the overall risk of bias for daily functioning in 10 studies, primarily due to the use of a self-reported outcome measure. One study had a high overall risk of bias, due to high risk in the domain of missing outcome data [145]. Due to the use of self-reported outcome measures and because fatigue outcomes may be unconsciously influenced in fatigue management studies due to an increased focus on the symptom [114], we rated eight studies as having a high risk of bias for domain 4 (measurement of the outcome). We rated one study as having some concerns and two studies at low risk of bias, one due to the use of an observer-based measurement [155] and the other due to the use of pseudo-blinding procedures [147].

Three RCTs that reported on the outcome of daily functioning were not included in the meta-analysis: one because the additional data was not received in time for our analysis, due to technical access issues [142], and the other two because the required data were not available in the published information [144, 150] (see [Table 2](#) for reasons for exclusion). We assessed the studies as having an overall high risk of bias, primarily due to the reliance on self-reported measures and the inability to timely analyse the delivered data [142] or to verify the reported information [144, 150].

Three non-randomised controlled studies that reported on the outcome of daily functioning were not included in the meta-analysis [152, 156, 157]. These studies were rated as having a critical risk of bias. Each study was judged to have either a critical risk of bias in at least one domain or a serious risk in several domains, leading to an overall judgment of a critical risk of bias according to the ROBINS-I V2 tool [162]. The main reasons for these judgments were due to poor descriptions or lack of information ([Table 3](#)).

Quality of life

Fourteen randomised controlled studies reported on the outcome quality of life [141, 142, 143, 144, 145, 146, 147, 149, 150, 151, 153, 154, 155, 159].

Twelve randomised controlled studies reporting on quality of life were included in the meta-analysis. We had some concerns about the overall risk of bias for quality of life in nine studies, as they scored some concerns in more than one domain of the RoB but did not show a high risk in any domain. Three studies were rated as having a high overall risk of bias due to a high risk in domain 3

(missing outcome data) in at least one of the time points [145, 150, 154].

Not all relevant data were available in a timely manner at the time of analysis. As a result, two randomised controlled trials reporting on quality of life outcomes could not be included in the meta-analysis: in one case, the published article did not provide sufficient relevant information [144], while in the other, additional data were supplied but were delayed due to technical access issues and thus not available in time for analysis [142]. Both studies were assessed as having an overall high risk of bias, primarily due to their reliance on self-reported measures and because the data could not be verified at the time of our analyses.

The non-randomised controlled studies included in the review did not report on quality of life.

Adverse events

It was not possible to formally assess the risk of bias for this outcome, as none of the studies included in the meta-analysis systematically reported adverse events. The studies that mentioned the absence of adverse events related to the intervention did not specify how these were assessed or monitored, nor did they report them per number of participants.

Participation

We identified three studies reporting on participation. We rated two studies as some concerns regarding the overall risk of bias, as both studies scored some concerns on domain 2 (deviations from intended interventions) due to the lack of blinding of the carers and the people delivering the interventions [149, 151]. Considering the nature of OT, no high risk was assigned to this domain. Additionally, we had some concerns in domain 4 (measurement of the outcome) due to the use of self-reported measures. We rated one study as having a high overall risk of bias due to a high risk in domain 5 (reporting bias) [159], as not all the subscales of the Impact on Participation and Autonomy (IPA) questionnaire were reported, potentially biasing the total score. We decided to include this study in the meta-analysis, as we used the SDM rather than the total score.

One RCT assessed relevant outcomes on participation but could not be included in the meta-analysis because the delivery of required additional information was delayed due to technical issues [142]. We assessed this study as having an overall high risk of bias, mainly due to the use of self-reported outcomes and because the data could not be incorporated in our analyses.

Self-efficacy

Eight studies reported on self-efficacy [143, 145, 146, 148, 150, 151, 153, 158]. Of these, we judged five as having some concerns overall because we had concerns in at least one domain of the RoB 2 tool but did not consider any to be at high risk of bias. We judged the other three studies to have an overall high risk of bias, primarily due to missing outcome data and selective reporting of the results.

Self-management

One study assessed self-management [155]. We had some concerns about its risk of bias due to the use of self-reported outcome measures.

Mood

Three studies reported on mood as an outcome [148, 153, 155]. We rated two of these studies as having some concerns regarding the risk of bias due to the use of self-reported measurements, while we judged one study to be at high risk of bias due to the randomisation process and missing outcome data.

Resilience

One study examined resilience [159]. We judged it as having some concerns due to the use of self-reported measurements.

Impact on caregivers

None of the included studies reported on the impact on caregivers.

Synthesis of results

The results of the meta-analyses for each outcome, based on three comparison groups (active control, usual care, no intervention) and three time points (post-intervention, medium-term follow-up, long-term follow-up), are shown in [Supplementary material 6](#) and summarised below, clustered by outcome and ordered by chronological time point and by comparison group. The forest plots include the pooled effects, as well as the effects with their 95% confidence intervals, risk of bias assessment, heterogeneity statistic, intervention type and the outcome measure used for each included study.

For each outcome, we provide a narrative synthesis of studies that we did not include in the meta-analyses. Six studies were excluded because relevant additional data could not be provided in a timely manner due to technical issues [142], data were not available [144, 150], or they were non-randomised studies with a serious or critical risk of bias [152, 156, 157]. These studies are summarised in [Table 2](#).

Critical outcomes

Daily functioning

Post-intervention, OT interventions may result in a small benefit for daily functioning when compared to active control (SMD 0.22, 95% CI -0.02 to 0.46; $I^2 = 17\%$; 7 studies, 364 participants; low certainty). We are very uncertain about the large effect observed in the OT groups compared to the usual care groups (SMD 1.19, 95% CI 0.29 to 2.09; $I^2 = 62\%$; 2 studies, 120 participants; very low certainty). Compared to no intervention, OT interventions may result in a moderate benefit post-intervention for daily functioning (SMD 0.56, 95% CI -0.26 to 1.37; $I^2 = 68\%$; 2 studies, 164 participants; low certainty) ([Summary of findings 1](#); [Summary of findings 3](#); [Summary of findings 5](#)).

At medium-term follow-up, OT interventions may have little to no effect on daily functioning compared to active control (SMD 0.11, 95% CI -0.13 to 0.35; $I^2 = 0\%$; 4 studies, 264 participants; low certainty), and a large effect compared to usual care (SMD 0.99, 95% CI 0.57 to 1.41; I^2 not applicable; 1 study, 98 participants; low certainty) ([Summary of findings 2](#); [Summary of findings 4](#)).

One study comparing the long-term effects of OT versus active control found that OT may have little to no effect on daily functioning, but the evidence is very uncertain (SMD -0.05, 95% CI -0.52 to 0.42; I^2 not applicable; 1 study, 69 participants; very low certainty). When compared to usual care, the evidence is very uncertain about the long-term effect of OT on daily functioning

(SMD 0.63, 95% CI 0.19 to 1.07; I^2 not applicable; 1 study, 86 participants; very low certainty) ([Supplementary material 10](#)).

No studies that were included in the meta-analyses reported medium- or long-term effects on any of the outcomes in the comparison of OT versus no intervention ([Summary of findings 6](#); [Supplementary material 10](#)).

Six studies were not included in the meta-analysis; four compared OT to usual care and two compared OT to no intervention. Results were either mixed or not reported in a useful way ([Table 2](#)).

Quality of life

Post-intervention, OT interventions may result in a small benefit compared to active control for mental health-related quality of life (HR-QoL) (SMD 0.33, 95% CI -0.01 to 0.66; $I^2 = 42\%$; 5 studies, 296 participants; low certainty), whereas little to no benefit was found for physical HR-QoL (SMD 0.17, 95% CI -0.09 to 0.42; $I^2 = 12\%$; 5 studies, 295 participants; low certainty). Compared to usual care, we are very uncertain whether OT interventions have a moderate effect on physical HR-QoL (SMD 0.69, 95% CI -1.18 to 2.56; $I^2 = 91\%$; 2 studies, 166 participants; very low certainty) and a small effect on mental HR-QoL (SMD 0.44, 95% CI -1.27 to 2.16; $I^2 = 91\%$; 2 studies, 166 participants; very low certainty), as these findings are based on very low certainty evidence. OT may result in a moderate post-intervention benefit when compared with no intervention on mental HR-QoL (SMD 0.68, 95% CI -0.09 to 1.45; $I^2 = 0\%$; 3 studies, 192 participants; very low certainty), but we are very uncertain about this evidence. We are also very uncertain about the results of OT versus no intervention for physical HR-QoL, which suggest little to no effect (SMD 0.12, 95% CI -0.62 to 0.86; $I^2 = 0\%$; 3 studies, 192 participants; very low certainty) ([Summary of findings 1](#); [Summary of findings 3](#); [Summary of findings 5](#)).

At medium-term follow-up, OT may result in a small benefit for mental HR-QoL (SMD 0.21, 95% CI -0.08 to 0.49; $I^2 = 20\%$; 4 studies, 270 participants; very low certainty) compared to active control, but we are very uncertain about this evidence. OT may have little to no effect on physical HR-QoL (SMD 0.19, 95% CI -0.05 to 0.43; $I^2 = 0\%$; 4 studies, 268 participants; low certainty), when compared to active control. Compared to usual care, we are very uncertain whether OT results in a small benefit for physical HR-QoL (SMD 0.26, 95% CI -0.33 to 0.86; $I^2 = 81\%$; 2 studies, 242 participants; very low certainty) and little to no difference in mental HR-QoL (SMD 0.18, 95% CI -0.63 to 0.98; $I^2 = 90\%$; 2 studies, 242 participants; very low certainty). No meta-analysis results are available for medium-term effects of OT compared to no intervention on quality of life ([Summary of findings 2](#); [Summary of findings 4](#); [Summary of findings 6](#)).

In the long term, one study comparing OT with active control found that OT may result in little to no long-term effect on physical HR-QoL (SMD 0.05, 95% CI -0.42 to 0.52; I^2 not applicable; 1 study, 69 participants; very low certainty) or mental HR-QoL (SMD 0.06, 95% CI -0.41 to 0.53; I^2 not applicable; 1 study, 69 participants; very low certainty), but the evidence is very low certainty. Compared to usual care, the evidence is very uncertain about the finding of little to no effect on physical HR-QoL (SMD 0.15, 95% CI -0.19 to 0.49; I^2 not applicable; 1 study, 131 participants; very low certainty) ([Supplementary material 10](#)).

Two studies not included in the meta-analysis reported mixed results on quality of life data (Table 2).

Adverse effects

No studies reported the frequency or type of any adverse effects in a systematic way. Eleven studies did not report any information on adverse events; eight studies reported, in a non-systematic way, that no adverse events occurred related to the intervention or study; and one study used a question at baseline, post-test and follow-up to measure adverse events. This study found no significant differences in falls, sickness, exacerbations, orthopaedic injuries or changes in medications across the study groups [151].

Important outcomes

We did not assess the certainty of the evidence for the important outcomes, other than for the outcome "participation".

Participation

Compared to active control, there may be little to no benefit of OT on participation post-intervention (SMD 0.18, 95% CI -0.18 to 0.53; $I^2 = 38\%$; 2 studies, 210 participants; low certainty); and with one study comparing OT to no intervention, we are very uncertain about the small effect observed for participation post-intervention (SMD 0.38, 95% CI -0.45 to 1.21; 23 participants; very low certainty). (Summary of findings 1; Summary of findings 5)

Two studies comparing OT with active control found little to no medium-term effect on participation (SMD 0.02, 95% CI -0.26 to 0.29; $I^2 = 0\%$; 2 studies, 210 participants; very low certainty) (Summary of findings 2).

The single study at long-term comparing OT with active control found little to no effect on participation (SMD -0.05, 95% CI -0.52 to 0.42; I^2 not applicable; 1 study, 69 participants; very low certainty), but we are very uncertain about this evidence (Supplementary material 10).

One study not included in the meta-analysis provided participation data. The study reported no differences between the intervention and control (usual care) groups on the reported IPA subscales [142] (Table 2).

Self-efficacy

Comparing OT with active control, little to no post-intervention benefit was found for self-efficacy (SMD 0.14, 95% CI -0.32 to 0.60; $I^2 = 47\%$; 3 studies, 198 participants); at medium-term follow-up, there were small benefits for self-efficacy (SMD 0.29, 95% CI -0.01 to 0.59; $I^2 = 0\%$; 2 studies, 170 participants).

OT may show small beneficial effects compared to usual care on self-efficacy post-intervention (SMD 0.33, 95% CI 0.07 to 0.58; $I^2 = 0\%$; 2 studies, 237 participants) and at long-term (SMD 0.24, 95% CI -0.10 to 0.58; I^2 not applicable; 1 study, 131 participants), but little to no effect was found at medium term (SMD 0.19, 95% CI -0.06 to 0.45; $I^2 = 0\%$; 2 studies, 242 participants).

The only study comparing self-efficacy following OT versus no intervention revealed little to no post-intervention effect of OT (SMD 0.09, 95% CI -0.14 to 0.32; 181 participants).

One study not included in the meta-analysis reported pre-post self-efficacy data of the intervention group only (Table 2).

Self-management

One study using active intervention as a comparator found a small post-intervention benefit for self-management in the OT group (SMD 0.36, 95% CI -0.31 to 1.03; 35 participants).

Mood

One study evaluated mood at post-intervention, comparing OT with active control and found moderate benefit for depression (SMD 0.63, 95% CI -0.05 to 1.32; I^2 not applicable; 1 study, 35 participants) and little to no effect on anxiety (SMD 0.17, 95% CI -0.50 to 0.83; I^2 not applicable; 1 study, 35 participants). Compared to usual care, OT may result in a small negative effect on depression post-intervention (SMD -0.21, 95% CI -0.53 to 0.12; I^2 not applicable; 1 study, 146 participants) and medium-term (SMD -0.43, 95% CI -0.99 to 0.13; $I^2 = 78\%$; 2 studies, 242 participants), and little to no effect on anxiety post-intervention (SMD -0.18, 95% CI -0.51 to 0.14; I^2 not applicable; 1 study, 146 participants) and medium-term (SMD -0.16, 95% CI -0.49 to 0.16; I^2 not applicable; 1 study, 144 participants).

No studies reported on long-term effects on mood.

Resilience

One study reporting on resilience found a small beneficial effect post-intervention when OT was compared to no intervention (SMD 0.22, 95% CI -0.61 to 1.04; 1 study, 23 participants).

Impact on caregivers

No studies reported on the impact on caregivers.

Sensitivity analyses

The forest plots of the sensitivity analyses for each comparison (i.e. excluding studies with overall high risk of bias) are available on request from the first author of the review.

Sensitivity analysis of comparison OT versus active control

After exclusion of the studies with an overall high risk of bias, the SMD of daily functioning at mid-term changed from little-to-no effect to small effect (SMD 0.24, 95% CI -0.07 to 0.54), and mental HR-QoL changed from a small benefit to little-to-no effect (SMD 0.14, 95% CI -0.63 to 0.91). The sensitivity analyses did not result in any differences for the other outcomes.

Sensitivity analysis of comparison OT versus usual care

We did not perform sensitivity analyses for this comparison as they were unnecessary for the following reasons: either the main analyses included only one study or both included studies were at high risk of bias or none of the included studies were at high risk of bias.

Sensitivity analysis of comparison OT versus no intervention

The sensitivity analyses did not result in differences in effects in any of the outcomes.

Equity assessment

We included three non-randomised studies in the review [152, 156, 157]. We reported data across all included studies, where available, using the PROGRESS-Plus framework [133] (Supplementary material 2). Most of the 20 included studies reported on location

($n = 20$), age ($n = 20$), sex ($n = 19$) and disability ($n = 14$), whereas fewer studies reported on race/ethnicity ($n = 7$), level of degree attainment ($n = 2$) [153, 158], number of years in education ($n = 2$) [143, 145] and socioeconomic status/occupation ($n = 10$). No data were reported on religion or social capital. [Supplementary material 9](#) summarises the characteristics of participants we would expect to see in the evidence and those actually represented in the included studies.

Of the available PROGRESS-Plus data, we evaluated imbalance at baseline and found most studies reported no significant baseline differences in demographic characteristics between groups ([Supplementary material 2](#)). Three studies noted differences in age, MS type or employment status, which were taken into account in the analyses [142, 147, 148]. The remaining studies did not assess or did not report outcome differences by equity-relevant characteristics (e.g. age, sex, race, socioeconomic status), highlighting a gap in equity-focused outcome analysis.

In one study, significant baseline differences were observed between the groups in terms of age ($P = 0.02$), PDDS score ($P = 0.001$) and employment ($P = 0.00$) [148]. These differences are particularly relevant because the intervention was delivered during standard working hours. This raises concerns about equitable access to the intervention, as working participants may have been unintentionally underrepresented in the intervention group.

While reasons for dropout were sometimes described, none of the studies systematically examined whether attrition was associated with equity-related characteristics. Therefore, there is insufficient evidence available to understand potential barriers to participation and retention in interventions for people with MS.

[Supplementary material 8](#) contains details of the Equity Checklist for Systematic Review Authors completed for this review [134].

DISCUSSION

Summary of main results

This review found that occupational therapy for people with multiple sclerosis (MS) may improve daily functioning and mental health-related quality of life post-intervention, regardless of the comparison intervention, with effects present at medium-term follow-up for mental health-related quality of life, daily functioning and physical health-related quality of life. Evidence on long-term effects is uncertain.

This review included 20 studies comprising 1628 adults with MS, most of them with low to moderate disability levels. Interventions primarily focused on fatigue management, skill training, self-management and cognitive support, with one study targeting social participation. Follow-up was typically post-intervention, with few studies including medium-term (seven studies) or long-term (two studies) assessments. Outcomes other than daily functioning and quality of life were rarely evaluated. Adverse events and caregiver impacts were largely unreported or inconsistently monitored, limiting insight into intervention safety.

Limitations of the evidence included in the review

The evidence identified in this review partially addresses the main objective of evaluating the benefits and harms of occupational therapy interventions in people with MS. While the included

studies consistently targeted daily functioning and quality of life, other important outcomes (e.g. participation) were rarely assessed. Furthermore, no studies systematically reported on potential harms or adverse events. This precludes firm conclusions being drawn about the safety profile of OT interventions. Regarding the secondary objectives of our review, the evidence was insufficient to determine whether the effects of OT differ by delivery format (individual versus group) or setting (outpatient, inpatient, home-based), as subgroup analyses could not be performed due to the limited number of studies per comparison category.

The certainty of the evidence was generally low, primarily due to downgrading due to methodological limitations and the indirectness of the OT interventions and methodological limitations across studies. A key challenge in OT research is the difficulty of implementing blinding procedures. Unlike pharmacological treatments, where blinding can be achieved through identical pills or placebos, OT interventions typically require active engagement, verbal instruction, and physical or cognitive activities that are difficult to mask. Moreover, most studies relied on self-report measures, which further impacts the certainty of the evidence.

In addition to these methodological concerns, the substantial heterogeneity in both the OT interventions and the comparators used across studies represents a significant limitation. The studies varied widely in their content, delivery format, intensity and targeted outcomes, reflecting the complex and multifaceted nature of OT practice in MS. This aggregation of diverse interventions under a single professional label inevitably results in low-certainty findings and small, difficult-to-interpret average effects [163, 164, 165].

During this review we noted other challenges in summarising the evidence on OT for MS. Below we explain the most important challenges related to OT outcomes and OT interventions.

Challenges in occupational therapy outcomes

The evaluation of the outcomes of OT interventions for individuals with MS presents significant challenges. These challenges arise primarily from the multidimensional nature of activity and participation outcomes, which are central to OT but difficult to measure comprehensively. Unlike more straightforward clinical assessments—such as dynamometry for muscle strength or $\text{VO}_2\text{-max}$ for aerobic capacity—OT activity and participation outcomes encompass a broad range of components, making them less amenable to standardised measurement.

For this review, we specifically focused on "daily functioning" and "quality of life" as primary outcomes, given their strong alignment with OT's overarching goal to support engagement in meaningful daily activities and participation. However, consistent with findings from previous studies [166, 167], our results indicate that most assessments employed in occupational therapy fall short of capturing the nuances of real-life activity and participation. There remains a tendency in both clinical practice and research to prioritise function outcomes, such as muscle strength or cognitive performance. While these measures are likely relevant as intermediate outcomes in clinical trials [168], they do not fully capture the lived experience or the daily functioning of individuals with chronic and variable conditions such as MS.

This gap highlights a critical need for a paradigm shift in how outcomes are measured in OT. Future research should prioritise activity-based and participation-focused outcome measures that better reflect the lived experiences of individuals with MS rather than instruments evaluating functions [167]. A potentially efficient way to assess health domains relevant to occupational therapy is using PROMIS (Patient-Reported Outcomes Measurement Information System), especially with computerised adaptive testing (CAT) [169]. PROMIS CATs can evaluate areas like impact of fatigue and pain, physical HR-QoL, and social participation—aligning well with client-centred goals [112]. Their standardised and efficient format make them valuable for both clinical and research settings. Besides, integrating the International Classification of Functioning, Disability and Health (ICF) framework can support this shift by providing a comprehensive model that links body functions and structures with activities and participation, all within the context of environmental and personal factors [23]. By adopting such multidimensional and person-centred approaches, it may be possible to more effectively demonstrate the impact that occupational therapy can have on the lives of individuals with MS.

Secondly, the subjective nature of self-reported outcome measures, while essential for capturing patient-centred perspectives, introduces variability and limitations in blinding that can lower the grading of evidence certainty in systematic reviews. While all included studies used self-reported instruments to evaluate at least one key outcome domain, only two incorporated observer-based assessments of daily functioning. Healthcare for individuals with MS is increasingly moving toward a person-centred approach, where patients are actively involved in their care by expressing their personal values and needs during clinical interactions. Patient-reported outcome measures (PROMs) play a key role in this shift, as they help shape individualised treatment goals and facilitate meaningful communication between patients and healthcare professionals [170].

Thirdly, while the standard Cochrane RoB tools are widely used in clinical research, their application in OT presents challenges. It does not fully account for the complexity, client-centred nature, and frequent use of self-reported outcomes typical of OT interventions. Most studies were rated as having a moderate to high risk of bias, mainly due to domain 2 (deviations from intended interventions) and domain 4 (outcome measurement). In OT, outcomes are often assessed using self-reported tools such as the Canadian Occupational Performance Measure (COPM) and Impact on Autonomy and Participation (IPA) [78, 171]. These tools reflect meaningful, client-centred outcomes but make blinding impossible, leading to high bias scores in domain 4.2. Similarly, therapists cannot be blinded to the interventions they deliver, affecting domain 2. This inherent limitation in OT practice can influence intervention delivery and bias ratings.

No studies were identified that examined the impact on caregivers. This may reflect a lack of research on this outcome within MS rehabilitation studies. Including and assessing the caregiver's perspective in both research and practice is important as MS affects not only the person diagnosed with MS but also their broader social and familial context [172]. Informal care plays a critical role in managing chronic conditions like MS, yet it imposes significant health, emotional and financial burdens on caregivers. While this unpaid input reduces healthcare costs for recipients and society,

caregivers often face reduced quality of life, increased loneliness and economic strain. In clinical practice, caregiver involvement often begins reactively, only once difficulties emerge. However, a more proactive approach that includes caregivers from the point of diagnosis may be more (cost-)effective in supporting people with MS and their caregivers [173].

Furthermore, most studies focused on short-term outcomes, with relatively few of them including medium- or long-term follow-up. Only a limited number of studies extended their evaluation beyond the outcomes of daily functioning and quality of life, underscoring the need for broader, more ecologically valid outcome measures in OT research.

These findings highlight the need for core outcome sets for OT practice and research in MS and more tailored appraisal tools that reflect the realities of OT interventions (and other behavioural interventions).

Challenges in occupational therapy interventions

Occupational therapy is inherently holistic, which complicates the attribution of observed outcomes to specific intervention components. Our review included a broad spectrum of specific OT interventions, such as fatigue management, skill training, self-management support and cognitive strategies. The overall effect of OT as a comprehensive intervention may therefore not align with the effects of isolated, targeted components. This perspective was reinforced by input from our partner with lived experience, who highlighted that OT should not be equated with single-component interventions. The heterogeneity in intervention content contributed to reduced confidence in the pooled effect estimates, leading us to downgrade the certainty of evidence where appropriate.

The included studies varied in the level of detail they provided about the occupational therapy interventions, which made direct comparison of their results challenging. To increase comparability, it would be helpful in future studies to use the International Classification of Health Interventions (ICHI) developed by the World Health Organisation to describe interventions according to the three axes of Target, Action and Means (<https://icd.who.int/dev11/l-ichi/en>), as well as the Template for Intervention Description and Replication (TIDieR) [109].

It is notable that while several studies compare interventions to 'usual care', only a few clearly define what 'usual care' entails. It is essential that studies clearly specify the components of 'usual care' as its content can vary widely in the context of MS. Access to rehabilitation services varies globally. For example, while approximately 75% of European countries report access to fatigue management interventions for people with MS, only 59% of countries globally provide such support [2]. Besides, a large international survey revealed that occupational therapists were involved in the care of just 16% of people with MS, with considerable variation between countries [174]. Tools such as the TIDieR checklist may improve reporting of both experimental and control interventions, including the specifics of usual care [109].

This review did not focus on mediators of the OT interventions. However, as prioritised in the Pathways to Cures, to enhance the effectiveness of non-pharmacological interventions such as OT, future research should aim to identify their active components and

underlying mechanisms of functional recovery, while also clarifying dose-response relationships and how different patient subgroups respond to treatment [175]. A realist review may provide insights from the available evidence of these processes [176].

Equity assessment

Globally, the number of occupational therapists working across countries in which data have been recorded (88 countries), ranges from 0.0002 to 16 per 100,000 head of population [131]. There is high unmet need in low-income countries and evidence of inequalities in access to health services for MS [130]. Substantial barriers exist for people with MS in accessing diagnosis, treatment and rehabilitation – with low-income countries having no access to high-efficacy licenced disease-modifying treatments and significantly less availability of MS rehabilitation [129]. Only one study in this review did not take place in a high-income country. The number of MS specialist neurologists across the included countries ranges (per 100,000 people) from 0.05 in Iran to 1.7 in Belgium; and MS nurses from 0.1 in Canada to 1.2 in Switzerland [129]. No data are available on the number of occupational therapists working with people with MS across the included study countries.

We developed a logic model prior to beginning the review to understand the potential impact of health equity on the intervention. The model describes what is known about health equity that may be important for the review. From the logic model, we know that not everyone who needs access to the intervention (occupational therapy) has access to it. We also know that even when it is available, not everyone will access the intervention for various reasons [177; 130].

Factors considered for the review can be seen in Figure 1. Not all factors considered in the logic model could be applied to the review as they are challenging to measure appropriately (e.g. health beliefs, culture, impact of public policy [177, 178]), but they remain important considerations for occupational therapy practice and were therefore included in the logic model. For example, the health policy context could impact the acceptability, appropriateness, availability and effectiveness of the occupational therapy intervention provided [177].

The studies in this review included adults with MS, ranging from 18 to 70 years old, most of them with mild to moderate levels of disability. The 3:1 ratio of women to men included in the studies was in line with the higher prevalence of females with MS [2]. Worldwide, about 62% of people with MS have mild to moderate disability levels, suggesting that occupational therapy was not studied in the 25% of people with MS with severe disability [2]. However, people with higher levels of disability may have increased needs for occupational therapy services, such as providing equipment to enable autonomy in personal care or adapting settings in their home, work or leisure context to reduce environmental barriers [32].

In summary, while most studies ensured demographic comparability, only a few addressed broader equity considerations such as cultural diversity, accessibility or representativeness. These gaps highlight the need for more inclusive research designs and reporting practice in future studies.

The absence of health equity-focused analysis from the included studies makes it difficult to predict if the intervention can be beneficial for specific subgroups equally; this should be further explored. Future research can potentially address this via the incorporation of frameworks such as the STEP UP and INCLUDE Ethnicity Framework, including equity-stratified analysis, and including detailed inclusive criteria targeting both the socially and clinically marginalised population [133, 179, 180]. Finally, integrating public and patient involvement (PPI) in all stages of a research trial may facilitate more inclusive and representative research [181].

Limitations of the review processes

To address the full spectrum of multidimensional consequences of MS, a multidisciplinary approach is preferred [182]. In line with our protocol, we included only those multidisciplinary programmes in which the role of occupational therapy was clearly defined and outcomes were distinctly reported. Notably, we excluded many studies due to the absence of a separate specification of the occupational therapist's role within a broader (multidisciplinary) intervention. However, in real-world practice, occupational therapists may work as part of a multidisciplinary team. Research exploring the unique contribution of occupational therapists within these teams remains both challenging and scarce [28, 183].

Future research could focus on exploring those types of interventions that align with the philosophy and approach of OT but were delivered by other professionals due to cultural or organisational constraints. An important consideration for future research is whether the focus should remain on specific professional disciplines, such as occupational therapy, or shift toward evaluating the intervention itself and its intended outcomes. Emphasising the effectiveness of therapeutic strategies—regardless of disciplinary boundaries—may offer a more comprehensive understanding of what benefits individuals with MS most, particularly when holistic and patient-centred goals are prioritised. This reflects a broader movement from multidisciplinary to transdisciplinary care, where collaboration transcends professional silos to address complex needs in an integrated way.

Fatigue is highly prevalent and substantially impacts daily functioning in people with MS [34]. As a result, fatigue is often targeted in behavioural interventions, with fatigue severity or frequency often used as an outcome measure [184]. In contrast, our review focused on broader outcomes related to activity and functioning. This broader approach provides a more holistic perspective on the impact of occupational therapy interventions in MS. However, this also led to the exclusion of several studies that only measured fatigue severity as an outcome, rendering them ineligible. Nonetheless, we included 10 studies evaluating the impact of fatigue on daily functioning.

Due to the lack of systematic data collection and unclear reporting, we were unable to extract data on adverse events. This gap in the evidence highlights an important area for improvement in future research. Specifically, in line with Consolidated Standards of Reporting Trials (CONSORT) Harms Extension Guidance [185], studies should incorporate clear and systematic approaches for assessing and reporting adverse events, regardless of their perceived relation to the intervention.

Agreements and disagreements with other studies or reviews

It is evident from this review that there has been an increase in the volume and quality of trials in occupational therapy and MS since the publication of the last Cochrane review on this topic [52]. We included 20 studies in this review, while the 2003 review included only three studies (one RCT). The increase in the quality and quantity of studies is encouraging, and the current review provides some support for the effectiveness of OT for people with MS. Research gaps remain regarding the effects and potential harms of occupational therapy in MS, and this review suggests that advancing the evidence base may require a rethinking of how OT trials are conceptualised and evaluated.

While some uncertainty persists regarding the magnitude of OT's effects, the direction of effect estimates tends to favour OT, although there are instances of benefit for the control group and several results that are equivocal. Effect sizes were generally larger when OT interventions were compared to no treatment as opposed to active control conditions. Therefore, although we downgraded the certainty of evidence in some cases due to heterogeneity and methodological limitations, the overall findings provide cautious but consistent support for the effectiveness of OT in MS.

There have been several recently published reviews relevant to OT for MS. In 2015, a Cochrane review of telerehabilitation for people with MS was published, which included interventions provided by occupational therapists [186]. From the nine RCTs included in the Cochrane review, limited evidence for the efficacy of the intervention was found for improving function, fatigue and quality of life. Of note, the search for this review was concluded in 2014 and huge strides have been made in telerehabilitation and online interventions since then, particularly since the onset of COVID-19 [187]. An update of this review is therefore warranted, as positive effects have been reported in other neurological conditions [188].

In 2019, a review of Cochrane reviews on rehabilitation for people with MS identified 15 reviews on MS rehabilitation but only one review of occupational therapy [36]. Moderate-quality evidence was found to support long-term gains of multidisciplinary rehabilitation programmes in improving activity and participation for people with MS. In this overview of Cochrane reviews (which included the 2007 review on the same topic [189]), occupational therapy was a named intervention in the review parameters. As a 'uni-disciplinary' approach (i.e. occupational therapy only), the authors report inconclusive evidence for function, participation and quality of life outcomes [36].

Quinn and Hynes (2020) conducted a scoping review of occupational therapy and MS, identifying 30 relevant records [53]. Their review included similar areas of occupational therapy practice, such as fatigue management, upper-limb rehabilitation, cognitive rehabilitation and others. The authors reported an improvement in the quality and quantity of RCTs included, although noted considerable variation in methodological design. While this review complements these results, it is important to highlight that Quinn and Hynes (2020) used broader inclusion criteria, not limiting their review to controlled trials [53]. Additionally, their results were reported narratively without conducting a meta-analysis.

We found one study reporting on economic analysis, which concluded that the cost-effectiveness data for the studied intervention (a fatigue management programme) were equivocal [153]. No evidence about cost-effectiveness was found or reported in the reviews discussed above.

AUTHORS' CONCLUSIONS

Implications for practice

The available evidence is based on 20 studies of occupational therapy, most of which were conducted in high-income countries and included participants with mild to moderate disability due to multiple sclerosis (MS).

In people with MS, occupational therapy (OT) may lead to small improvements in daily functioning and mental aspects of quality of life immediately after the intervention, when compared to active control or no intervention. Compared to usual care, OT may result in moderate improvements in daily functioning, but the evidence is very uncertain. Occupational therapy may have little or no effect on physical aspects of quality of life or participation, regardless of the comparator.

At medium-term follow-up (up to six months), OT may continue to offer small benefits for the mental aspects of quality of life when compared to active control. Compared to usual care, OT may result in moderate improvements in daily functioning in the medium term, although the evidence remains limited. However, there is little to no evidence of benefit for physical quality of life or participation at this time point, and the certainty of the evidence is low to very low.

Due to the low to very low certainty of evidence available for OT interventions for people with multiple sclerosis, we could not draw firm conclusions about their long-term effects on daily functioning, quality of life or participation.

Our pooled effects require cautious interpretation because the certainty we can have about the evidence is limited by the heterogeneity of the interventions and our exclusion of multidisciplinary studies that did not have separate data for the impact of OT specifically.

No studies systematically reported adverse effects. Therefore, we are uncertain about the safety profile of OT interventions. The results of this review can only be applied to adults with multiple sclerosis who have mild to moderate disability and who receive care in high-income countries; the evidence does not allow us to reach conclusions about the effects of OT for people with more severe disability or people in low-resource settings.

While OT may offer short-term benefits for daily functioning and the mental aspects of quality of life, expectations should be managed regarding its impact on physical quality of life and social participation. Decisions about referral and intervention planning should be tailored to individual patient goals, preferences and context, and should take into account the limited evidence on long-term outcomes and safety.

Equity-related implications for practice

The studies included in this review suggest that although OT interventions may generally be effective, we are not sure whether

the benefits are evenly distributed among all participants with MS. That is, the evidence is limited as most studies did not perform subgroup analysis to further investigate effectiveness. Although several studies include indicators of potential disadvantage, such as unemployment, high levels of disability and severity of symptoms, few studies have examined these factors in relation to how they may influence intervention outcomes. Therefore, we do not have equity-focused interpretations of the results.

Implications for research

Due to the low to very low certainty evidence available for OT interventions in people with MS, further high-quality research is needed to clarify their effects post-intervention and in the medium and long term, and across diverse populations and settings.

Future studies should prioritise outcomes that reflect the goals of OT, including daily functioning, quality of life and participation. These outcomes should be measured using validated tools that capture real-world performance and patient-centred changes. The use of standardised outcome sets, such as those aligned with the International Classification of Functioning, Disability and Health, is encouraged. Longer follow-up periods are needed to assess the sustainability of effects. Additionally, adverse events should be systematically monitored and reported using established guidelines such as the CONSORT Harms Extension [185].

The interventions studied varied widely in content, intensity and delivery format. Future trials should clearly define the OT components, using standardised frameworks such as the Template for Intervention Description and Replication (TIDieR) [109] and the International Classification of Health Interventions [190]. Research should also examine the effects of different delivery formats (e.g. individual versus group, in-person versus telehealth, occupational therapy as part of multidisciplinary care or as a single discipline), and variations in the frequency and duration of therapy, in order to determine optimal intervention strategies.

Comparators in future studies should be clearly described. When using 'usual care' as a comparator, researchers should specify what this entails to improve interpretability and reproducibility. Direct comparisons between OT and other active interventions (e.g. physiotherapy, cognitive rehabilitation) would also be valuable to consider the comparative effects of this therapeutic approach.

Most studies included in this review focused on adults with mild to moderate disability living in high-income countries. Future research should include people with more severe disability, older adults and those from underrepresented or underserved populations, including individuals in low- and middle-income countries. Studies should also explore the impact of OT in different clinical settings, such as inpatient rehabilitation and community-based care.

The current evidence base is limited by methodological weaknesses, including high risk of bias due to lack of blinding, small sample sizes and inconsistent reporting. Future trials should use robust randomisation procedures, blinded outcome assessment where feasible, and transparent reporting of all prespecified outcomes. Larger, multicentre trials are needed to improve precision and generalisability. Researchers could use the new CONSORT 2025 statement as a guiding framework when designing and reporting randomised controlled trials, as it enhances

methodological transparency, facilitates critical appraisal, and increases the reliability and impact of their findings [191].

Equity-related implications for research

The absence of health equity-focused analyses limits our understanding of whether OT interventions benefit all subgroups equally. Equity-focused research is therefore needed. Future studies should report and analyse outcomes by relevant subgroups (e.g. age, sex, socioeconomic status, disability level) and consider barriers to access and participation. Applying frameworks such as PROGRESS-Plus [133], STEP UP [179] and the INCLUDE Ethnicity Framework [192], as well as including people with lived experience in the research team, would support more inclusive and representative research. Ultimately, this approach will help lead to findings that are both relevant and fair for all individuals living with MS.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD015371.pub2](https://doi.org/10.1002/14651858.CD015371.pub2).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Equity checklist

Supplementary material 9 Summary of the characteristics of participants we would expect to see in the evidence and the actual participant characteristics extracted from the included studies

Supplementary material 10 Summary of findings tables of long-term effects of occupational therapy compared to three comparison subgroups (active control, usual care, no intervention)

ADDITIONAL INFORMATION

Acknowledgements

We are grateful to Esther Steultjens (previously affiliated to the Institute of Health Studies, HAN University of Applied Sciences, Nijmegen, the Netherlands) and Rebecca Morisson (expert MS) for their assistance in protocol preparation. The authors wish to thank Thomas Vandendriessche, Eline Vancoppenolle, and Krizia Tuand, the biomedical reference librarians of the KU Leuven Libraries – 2Bergen – learning Centre Désiré Collen (Leuven, Belgium), for their help in conducting the systematic literature search.

We thank Inger Grethe Loning and Sofie Ferdinand, chairs of the Special Interest Group Occupation of the Rehabilitation in Multiple Sclerosis (RIMS) network for their continuous support of the project and valuable feedback. We thank Jessica Sayfullaeva (research student at University of Galway, Ireland) for support with the equity assessment.

Editorial and peer-reviewer contributions

Cochrane Multiple Sclerosis and Rare Diseases of the Central Nervous System Review Group supported the authors in the development of this focused review.

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Graziella Filippini, Fondazione IRCCS, Neurologico Carlo Besta, Milan, Italy
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Sue Marcus, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, selected peer reviewers, collated peer-reviewer comments and supported the editorial team): Cynthia Stafford, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer-reviewers (provided comments and recommended an editorial decision): Chiara Arienti, Clinical Epidemiology Research Center (CERC), Department of Biomedical Sciences, Humanitas University, Pieve Emanuele, Milan, Italy (clinical review); Paul McKeegan, MS patient (patient and public review); Dr Kerry Dwan, University of York (methods review); Jo Platt, Central Editorial Information Specialist (search review). One additional peer reviewer provided clinical peer review but chose not to be publicly acknowledged.

The following people conducted the editorial process for the protocol.

- Sign-off Editor (final editorial decision): Robert Boyle, Cochrane's Editorial Board, Imperial College London, UK
- Managing Editor (selected peer reviewers, collated peer-reviewer comments, provided editorial guidance to authors, edited the article): Samuel Hinsley, Central Editorial Service
- Editorial Assistant (conducted editorial policy checks and supported the editorial team): Leticia Rodrigues, Central Editorial Service
- Copy Editor (copy editing and production): Lisa Winer
- Peer reviewers of the protocol (provided comments and recommended an editorial decision): Cara L Brown, College of Rehabilitation Sciences, University of Manitoba (clinical review); Herbert Karparkin, Physical Therapy Department, Hunter College (clinical review); Iván Pérez-Neri, National Institute of Neurology and Neurosurgery Manuel Velasco Suárez (consumer review); Assoc Prof Vanessa Jordan (methods review). One additional peer reviewer provided search peer review but chose not to be publicly acknowledged.

Contributions of authors

Conceptualisation: Daphne Kos, Isaline Eijssen, Sinéad Hynes, Geertruida Bekkering

Methodology: development of the strategy to search and select studies for inclusion in the review: Geertruida Bekkering, Leen De Coninck

Search and selection of studies for inclusion in the review: Ashley Boers, Ciara O'Meara, Leen De Coninck

Collection of data for the review: Ashley Boers, Ciara O'Meara, Jennifer Freeman, Isaline Eijssen, Daphne Kos

Assessment of the risk of bias in the included studies: Ashley Boers, Leen De Coninck

Assessment of the certainty in the body of evidence: Ashley Boers, Geertruida Bekkering, Leen De Coninck

Analysis: Geertruida Bekkering, Ashley Boers, Ciara O'Meara, Daphne Kos

Interpretation of data: Daphne Kos, Ashley Boers, Geertruida Bekkering, Leen De Coninck, Ciara O'Meara, Marja Koen, Sinéad Hynes, Jennifer Freeman, Isaline Eijssen

Validation (ensuring the accuracy, precision, reliability, quality, and the overall integrity of the review): Geertruida Bekkering, Leen De Coninck, Jennifer Freeman, Daphne Kos

Writing - original draft of the review: Daphne Kos, Ashley Boers, Ciara O'Meara, Sinéad Hynes, Geertruida Bekkering, Jennifer Freeman, Isaline Eijssen

Writing - checking drafts and editing the review: Daphne Kos, Ashley Boers, Geertruida Bekkering, Leen De Coninck, Ciara O'Meara, Marja Koen, Sinéad Hynes, Jennifer Freeman, Isaline Eijssen

Project administration and supervision: Daphne Kos

Declarations of interest

Daphne Kos is the first author of a study included in the review [147]. DK declares that she has no commercial conflicts of interest.

Ashley Boers works as an implementation scientist and declares that she has no conflicts of interest.

Ciara O'Meara holds a health professional licence and declares that she has no conflicts of interest.

Geertruida E Bekkering declares that she has no conflicts of interest.

Leen De Coninck declares that she has no conflicts of interest.

Marja Koen declares that she has no conflicts of interest.

Jennifer Freeman works as clinical specialist physiotherapist and declares that she has no conflicts of interest.

Sinéad M Hynes is the senior author of a study included in the review [146]. SH holds a health professional licence and declares that she has no commercial conflicts of interest.

Isaline CJM Eijssen is the first author of a study included in the review [142]. ICJME works as a health professional and declares that she has no commercial conflicts of interest.

None of the review authors who are authors of eligible studies were involved in the development of the review protocol sections related to search methods and selection of studies. Furthermore, they were not involved in selection of studies, data extraction, risk of bias assessment or GRADE assessment of studies on which they were authors.

Sources of support

Internal sources

- KU Leuven & National MS Center Melsbroek, Belgium

Salary for Daphne Kos

- Cochrane Belgium & KU Leuven, Belgium

Salary for Geertruida E Bekkering

- Vrije Universiteit Amsterdam, Amsterdam UMC, Amsterdam, Netherlands

Salary for Isaline CJM Eijssen

- University of Ireland Galway, Ireland

Salary for Sinéad Hynes and Ciara O'Meara

- University of Plymouth, UK

Salary for Jennifer Freeman

External sources

- Elizabeth Casson Trust, UK

Research Grant 2022

- Evidence Synthesis Ireland, Ireland

History

Protocol first published: Issue 5, 2023

Ciara O'Meara was part supported by the Health Research Board (Ireland) and the Health and Social Care (HSC) Public Health Agency (Grant Number ESI-2021-001) through Evidence Synthesis Ireland/Cochrane Ireland

Registration and protocol

Protocol (2023) DOI 10.1002/14651858.CD015371 [60]

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library: full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies (included or excluded during the full-text screen) in the final review ([Supplementary material 2](#); [Supplementary material 3](#)); study data ([Supplementary material 7](#)), including study information, study arms, and study results or test data; consensus risk of bias assessments ([Supplementary material 5](#)); and analysis data, including overall estimates and settings and individual data rows ([Supplementary material 6](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan [108], using the inbuilt computation methods. Template Excel data extraction forms are available from the authors on reasonable request.

REFERENCES

1. Barten LJ, Allington DR, Procacci KA, Rivey MP. New approaches in the management of multiple sclerosis. *Drug Design, Development and Therapy* 2010;**4**:343-66. [DOI: [10.2147/DDDT.S9331](https://doi.org/10.2147/DDDT.S9331)]
2. MS International Federation MFIS. Atlas of MS. www.atlasofms.org 2020 (accessed 30 May 2025).
3. Walton C, King R, Rechtman L, Kaye W, Leray E, Marrie RA, et al. Rising prevalence of multiple sclerosis worldwide: insights from the Atlas of MS. *Multiple Sclerosis Journal* 2020;**26**(14):1816-21. [DOI: [10.1177/1352458520970841](https://doi.org/10.1177/1352458520970841)]
4. Wijeratne T, Grisold W, Baneke P, Helme A, King R, Carroll W. World Brain Day 2021: a call to stop multiple sclerosis. *Lancet Neurology* 2021;**20**(8):597-8. [DOI: [10.1016/S1474-4422\(21\)00200-3](https://doi.org/10.1016/S1474-4422(21)00200-3)]
5. Klineova S, Lublin FD. Clinical course of multiple sclerosis. *Cold Spring Harbor Perspectives in Medicine* 2018;**8**(9):a028928. [DOI: [10.1101/cshperspect.a028928](https://doi.org/10.1101/cshperspect.a028928)]
6. Scalfari A, Knappertz V, Cutter G, Goodin DS, Ashton R, Ebers GC. Mortality in patients with multiple sclerosis. *Neurology* 2013;**81**(2):184-92. [DOI: [10.1212/WNL.0b013e31829a3388](https://doi.org/10.1212/WNL.0b013e31829a3388)]
7. Oh J, Vidal-Jordana A, Montalban X. Multiple sclerosis: clinical aspects. *Current Opinion in Neurology* 2018;**31**(6):752-9. [DOI: [10.1097/WCO.0000000000000622](https://doi.org/10.1097/WCO.0000000000000622)]
8. Smyrke N, Dunn N, Murley C, Mason D. Standardized mortality ratios in multiple sclerosis: systematic review with meta-analysis. *Acta Neurologica Scandinavica* 2022;**145**:360-70. [DOI: [10.1111/ane.13559](https://doi.org/10.1111/ane.13559)]
9. Koch-Henriksen N, Laursen B, Stenager E, Magyari M. Excess mortality among patients with multiple sclerosis in Denmark has dropped significantly over the past six decades: a population based study. *Journal of Neurology, Neurosurgery, and Psychiatry* 2017;**88**(8):626-31. [DOI: [10.1136/jnnp-2017-315907](https://doi.org/10.1136/jnnp-2017-315907)]
10. Dobson R, Giovannoni G. Multiple sclerosis - a review. *European Journal of Neurology* 2019;**26**(1):27-40. [DOI: [10.1111/ene.13819](https://doi.org/10.1111/ene.13819)]
11. Feinstein A, Freeman J, Lo AC. Treatment of progressive multiple sclerosis: what works, what does not, and what is needed. *Lancet Neurology* 2015;**14**(2):194-207. [DOI: [10.1016/S1474-4422\(14\)70231-5](https://doi.org/10.1016/S1474-4422(14)70231-5)]
12. Kesselring J, Beer S. Symptomatic therapy and neurorehabilitation in multiple sclerosis. *Lancet Neurology* 2005;**4**(10):643-52. [DOI: [10.1016/S1474-4422\(05\)70193-9](https://doi.org/10.1016/S1474-4422(05)70193-9)]
13. Beckerman H, Kempen JC, Knol DL, Polman CH, Lankhorst GJ, de Groot V. The first 10 years with multiple sclerosis: the longitudinal course of daily functioning. *Journal of Rehabilitation Medicine* 2013;**45**(1):68-75. [DOI: [10.2340/16501977-1079](https://doi.org/10.2340/16501977-1079)]
14. Boeschoten RE, Braamse AM, Beekman AT, Cuijpers P, Van Oppen P, Dekker J, et al. Prevalence of depression and anxiety in multiple sclerosis: a systematic review and meta-analysis. *Journal of the Neurological Sciences* 2017;**372**:331-41. [DOI: [10.1016/j.jns.2016.11.067](https://doi.org/10.1016/j.jns.2016.11.067)]
15. Koch MW, Uyttenboogaart M, Van Harten A, Heerings M, De Keyser J. Fatigue, depression and progression in multiple sclerosis. *Multiple Sclerosis Journal* 2008;**14**(6):815-22. [DOI: [10.1177/1352458508088937](https://doi.org/10.1177/1352458508088937)]
16. Feinstein A, Magalhaes S, Richard JF, Audet B, Moore C. The link between multiple sclerosis and depression. *Nature Reviews Neurology* 2014;**10**(9):507-17. [DOI: [10.1038/nrneurol.2014.139](https://doi.org/10.1038/nrneurol.2014.139)]
17. Morrow SA, Rosehart H, Pantazopoulos K. Anxiety and depressive symptoms are associated with worse performance on objective cognitive tests in MS. *Journal of Neuropsychiatry and Clinical Neurosciences* 2016;**28**(2):118-23. [DOI: [10.1176/appi.neuropsych.15070167](https://doi.org/10.1176/appi.neuropsych.15070167)]
18. Nourbakhsh B, Julian L, Waubant E. Fatigue and depression predict quality of life in patients with early multiple sclerosis: a longitudinal study. *European Journal of Neurology* 2016;**23**(9):1482-6. [DOI: [10.1111/ene.13102](https://doi.org/10.1111/ene.13102)]
19. Chipchase SY, Lincoln NB. Factors associated with carer strain in carers of people with multiple sclerosis. *Disability and Rehabilitation* 2001;**23**(17):768-76. [DOI: [10.1080/09638280110062158](https://doi.org/10.1080/09638280110062158)]
20. Giordano A, Ferrari G, Radice D, Randi G, Bisanti L, Solari A; POSMOS study. Health-related quality of life and depressive symptoms in significant others of people with multiple sclerosis: a community study. *European Journal of Neurology* 2012;**19**(6):847-54. [DOI: [10.1111/j.1468-1331.2011.03638.x](https://doi.org/10.1111/j.1468-1331.2011.03638.x)]
21. Green R, Cutter G, Friendly M, Kister I. Which symptoms contribute the most to patients' perception of health in multiple sclerosis? *Multiple Sclerosis Journal - Experimental, Translational and Clinical* 2017;**3**(3):2055217317728301. [DOI: [10.1177/2055217317728301](https://doi.org/10.1177/2055217317728301)]
22. Kratz AL, Braley TJ, Foxen-Craft E, Scott E, Murphy III JF, Murphy SL. How do pain, fatigue, depressive, and cognitive symptoms relate to well-being and social and physical functioning in the daily lives of individuals with multiple sclerosis? *Archives of Physical Medicine and Rehabilitation* 2017;**98**(11):2160-6. [DOI: [10.1016/j.apmr.2017.07.004](https://doi.org/10.1016/j.apmr.2017.07.004)]
23. World Health Organization. International Classification of Functioning, Disability and Health (ICF). <https://www.who.int/standards/classifications/international-classification-of-functioning-disability-and-health-2001> (accessed May 2025).
24. Law M. Participation in the occupations of everyday life. *The American journal of occupational therapy* 2002;**56**(6):640-649. [DOI: [10.5014/ajot.56.6.640](https://doi.org/10.5014/ajot.56.6.640)]
25. McCabe MP, McKern S, McDonald E. Coping and psychological adjustment among people with multiple

sclerosis. *Journal of Psychosomatic Research* 2004;**56**(3):355-61. [DOI: [10.1016/S0022-3999\(03\)00132-6](https://doi.org/10.1016/S0022-3999(03)00132-6)]

26. Conradsson D, Ytterberg C, Von Koch L, Johansson S. Changes in disability in people with multiple sclerosis: a 10-year prospective study. *Journal of Neurology* 2018;**265**(1):119-26. [DOI: [10.1007/s00415-017-8676-8](https://doi.org/10.1007/s00415-017-8676-8)]

27. Jansa J, Ferdinand S, Milo M, Løyning IG, Huilla T, Kallmayer L, et al. Performance of activities of daily living in people with multiple sclerosis. *Multiple Sclerosis and Related Disorders* 2022;**57**:103342. [DOI: [10.1016/j.msard.2021.103342](https://doi.org/10.1016/j.msard.2021.103342)]

28. Momsen A-MH, Ørtenblad L, Maribo T. Effective rehabilitation interventions and participation among people with multiple sclerosis: an overview of reviews. *Annals of Physical and Rehabilitation Medicine* 2022;**65**(1):101529. [DOI: [10.1016/j.rehab.2021.101529](https://doi.org/10.1016/j.rehab.2021.101529)]

29. Strupp J, Romotzky V, Galushko M, Golla H, Voltz R. Palliative care for severely affected patients with multiple sclerosis: when and why? Results of a Delphi survey of health care professionals. *Journal of Palliative Medicine* 2014;**17**(10):1128-36. [DOI: [10.1089/jpm.2013.0667](https://doi.org/10.1089/jpm.2013.0667)]

30. Chiu C, Park M, Hoffman T, Campbell M, Bishop M. Descriptive analysis of free-text comments on healthcare priorities and experiences in a national sample of people with multiple sclerosis. *Multiple Sclerosis and Related Disorders* 2019;**34**:141-9. [DOI: [10.1016/j.msard.2019.06.023](https://doi.org/10.1016/j.msard.2019.06.023)]

31. Rommer PS, Eichstadt K, Ellenberger D, Flachenecker P, Friede T, Haas J, et al. Symptomatology and symptomatic treatment in multiple sclerosis: results from a nationwide MS registry. *Multiple Sclerosis Journal* 2019;**25**(12):1641-52. [DOI: [10.1177/1352458518799580](https://doi.org/10.1177/1352458518799580)]

32. World Federation of Occupational Therapists (WFOT). About Occupational Therapy. <https://wfot.org/about/about-occupational-therapy> (accessed April 2022).

33. World Federation of Occupational Therapists (WFOT). About Occupational Therapy. <https://wfot.org/news/2025/updated-definition-of-occupational-therapy> (Accessed Dec 24 2025) 2025.

34. Finlayson M. Multiple Sclerosis Rehabilitation: from Impairment to Participation. Boca Raton, FL: Taylor and Francis, 2012.

35. Taylor RR, Lee SW, Kielhofner G, Ketkar M. Therapeutic use of self: a nationwide survey of practitioners' attitudes and experiences. *American Journal of Occupational Therapy*. 2009;**63**(2):198-207. [DOI: [10.5014/ajot.63.2.198](https://doi.org/10.5014/ajot.63.2.198)]

36. Amatya B, Khan F, Galea M. Rehabilitation for people with multiple sclerosis: an overview of Cochrane Reviews. *Cochrane Database of Systematic Reviews* 2019, Issue 1. Art. No: CD012732. [DOI: [10.1002/14651858.CD012732.pub2](https://doi.org/10.1002/14651858.CD012732.pub2)]

37. Cason J. Telehealth: a rapidly developing service delivery model for occupational therapy. *International Journal of Telerehabilitation* 2014;**6**(1):29-35. [DOI: [10.5195/ijt.2014.6148](https://doi.org/10.5195/ijt.2014.6148)]

38. American Occupational Therapy Association AOTA. Occupational therapy's perspective on the use of environments and contexts to support health and participation in occupations. *American Journal of Occupational Therapy* 2016;**64**(6 (Supplement)):S57-S69. [DOI: [10.5014/ajot.2010.64S57](https://doi.org/10.5014/ajot.2010.64S57)]

39. Jesus TS, Landry MD, Hoenig H, Dussault G, Koh GC, Fronteira I. Is physical rehabilitation need associated with the rehabilitation workforce supply? *International Journal of Health Policy and Management* 2022;**11**(4):434-42. [DOI: [10.34172/IJHPM.2020.150](https://doi.org/10.34172/IJHPM.2020.150)]

40. World Federation of Occupational Therapists. Occupational Therapy Human Resources Project 2020 - Numerical. www.wfot.org/resources/occupational-therapy-human-resources-project-2020-numerical May 2020 (accessed January 2026).

41. Martinková P, Freeman J, Drabinová A, Erosheva E, Cattaneo D, Jonsdottir J, et al. Physiotherapeutic interventions in multiple sclerosis across Europe: Regions and other factors that matter. *Multiple Sclerosis and Related Disorders* 2018;**22**:59-67. [DOI: [10.1016/j.msard.2018.03.005](https://doi.org/10.1016/j.msard.2018.03.005)]

42. American Occupational Therapy Association. Telehealth in occupational therapy [position paper]. *American Journal of Occupational Therapy*. 2018;**72**(Supplemental 2):7212410059p1-7212410059p18. [DOI: [10.5014/ajot.2018.72S219](https://doi.org/10.5014/ajot.2018.72S219)]

43. American Occupational Therapy Association. Telehealth [position paper]. *American Journal of Occupational Therapy*. 2013;**67**(Suppl):S69-S90. [DOI: [10.5014/ajot.2013.67S69](https://doi.org/10.5014/ajot.2013.67S69)]

44. Little LM, Pickett KA, Proffitt R, Cason J. Keeping pace with 21st century healthcare: a framework for telehealth research, practice, and program evaluation in occupational therapy. *International Journal of Telerehabilitation* 2021;**13**(1):e6379. [DOI: [10.5195/ijt.2021.6379](https://doi.org/10.5195/ijt.2021.6379)]

45. World Federation of Occupational Therapists (WFOT). World Federation of Occupational Therapists' position statement on telehealth. *International Journal of Telerehabilitation* 2014;**6**(1):37-9. [DOI: [10.5195/IJT.2014.6153](https://doi.org/10.5195/IJT.2014.6153)]

46. Aterman S, Ghahari S, Kessler D. Characteristics of peer-based interventions for individuals with neurological conditions: a scoping review. *Disability and Rehabilitation* 2022;**Jan 27**:1-32. [DOI: [10.1080/09638288.2022.2028911](https://doi.org/10.1080/09638288.2022.2028911)]

47. Pentland D, Kantartzis S, Clausen MG, Wittemyre K. Occupational Therapy and Complexity: Defining and Describing Practice. London: Royal College of Occupational Therapists, 2018. [ISBN: 978-1-905944-71-2] [RCOT: www.rcot.co.uk]

48. Christiansen CH, Baum CM, Bass JD. Occupational Therapy: Performance, Participation, and Well-Being. 4th edition. Thorofare, NJ: Slack Incorporated, 2015.

49. Celani MG, Nonino F, Mahan K, Orso M, Ridley B, Baldin E, et al. Identifying unanswered questions and setting the agenda for future systematic research in multiple sclerosis. A worldwide, multi-stakeholder priority setting project. *Multiple*

Sclerosis and Related Disorders 2022;**60**:103688. [DOI: [10.1016/j.msard.2022.103688](https://doi.org/10.1016/j.msard.2022.103688)]

50. National Institute for Health and Care Excellence (NICE). Multiple sclerosis in adults: management - clinical guideline [CG186]. <https://www.nice.org.uk/guidance/cg186> 2019 (accessed 23 March 2022).

51. James Lind Alliance. Top 10 Research Priorities for Occupational Therapy. <https://www.jla.nihr.ac.uk/priority-setting-partnerships/occupational-therapy#tab-27806> 2020.

52. Steultjens EM, Dekker J, Bouter LM, Cardol M, Van de Nes JC, Van den Ende CH. Occupational therapy for multiple sclerosis. *Cochrane Database of Systematic Reviews* 2003, Issue 3. Art. No: CD003608. [DOI: [10.1002/14651858.CD003608](https://doi.org/10.1002/14651858.CD003608)]

53. Quinn É, Hynes SM. Occupational therapy interventions for multiple sclerosis: a scoping review. *Scandinavian Journal of Occupational Therapy* 2021;**28**(5):399-414. [DOI: [10.1080/11038128.2020.1786160](https://doi.org/10.1080/11038128.2020.1786160)]

54. De-Bernardi-Ojuel L, Torres-Collado L, García-de-la-Hera M. Occupational therapy interventions in adults with multiple sclerosis or amyotrophic lateral sclerosis: a scoping review. *International Journal of Environmental Research and Public Health* 2021;**18**(4):1432. [DOI: [10.3390/ijerph18041432](https://doi.org/10.3390/ijerph18041432)]

55. Yu CH, Mathiowetz V. Systematic review of occupational therapy-related interventions for people with multiple sclerosis: part 1. Activity and participation. *American Journal of Occupational Therapy*. 2014;**68**(1):27-32. [DOI: [10.5014/ajot.2014.008672](https://doi.org/10.5014/ajot.2014.008672)]

56. Yu CH, Mathiowetz V. Systematic review of occupational therapy-related interventions for people with multiple sclerosis: part 2. Impairment. *American Journal of Occupational Therapy*. 2014;**68**(1):33-8. [DOI: [10.5014/ajot.2014.008680](https://doi.org/10.5014/ajot.2014.008680)]

57. Finlayson M, Garcia JD, Preissner K. Development of an educational programme for caregivers of people aging with multiple sclerosis. *Occupational Therapy International* 2008;**15**(1):4-17. [DOI: [10.1002/oti.243](https://doi.org/10.1002/oti.243)]

58. Finlayson M, Preissner K, Garcia J. Pilot study of an educational programme for caregivers of people ageing with multiple sclerosis. *British Journal of Occupational Therapy* 2009;**72**(1):11-9. [DOI: [10.1177/030802260907200](https://doi.org/10.1177/030802260907200)]

59. Ortiz-Rubio A, Cabrera-Martos I, Rodriguez-Torres J, Fajardo-Contreras W, Diaz-Pelegrina A, Valenza MC. Effects of a home-based upper limb training program in patients with multiple sclerosis: a randomized controlled trial. *Archives of Physical Medicine and Rehabilitation* 2016;**97**(12):2027-33. [DOI: [10.1016/j.apmr.2016.05.018](https://doi.org/10.1016/j.apmr.2016.05.018)]

60. Kos D, Eijssen I, Bekkering GE, De Coninck L, O'Meara C, Koen M, et al. Occupational therapy in multiple sclerosis. *Cochrane Database of Systematic Reviews* 2023, Issue 5. Art. No: CD015371. [DOI: [10.1002/14651858.CD015371](https://doi.org/10.1002/14651858.CD015371)]

61. Bramer WM, Giustini D, De Jonge GB, Holland L, Bekhuis T. De-duplication of database search results for systematic reviews in EndNote. *Journal of the Medical Library Association*

2016;**104**(3):240-3 (correction published in *Journal of the Medical Library Association*, 105(1):111. doi: 10.5195/jmla.2017.128). [DOI: [10.3163/1536-5050.104.3.014](https://doi.org/10.3163/1536-5050.104.3.014)]

62. Covidence. Version accessed 20 April 2023. Melbourne, Australia: Veritas Health Innovation, 2023. Available at <https://www.covidence.org>.

63. Reeves BC, Deeks JJ, Higgins JP, Shea B, Tugwell P, Wells GA. Chapter 24: Including non-randomized studies on intervention effects. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s).. In: *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3>. Cochrane, 2022.

64. Tugwell P, Petticrew M, Kristjansson E, Welch V, Ueffing E, Waters E, et al. Assessing equity in systematic reviews: realising the recommendations of the Commission on Social Determinants of Health. *BMJ (Clinical Research Ed.)* 2010;**341**:c4739. [DOI: [10.1136/bmj.c4739](https://doi.org/10.1136/bmj.c4739)]

65. World Federation of Occupational Therapists (WFOT). Definitions of occupational therapy from member organisations. <https://wfot.org/resources/definitions-of-occupational-therapy-from-member-organisations> (accessed January 2026) 2018.

66. Daniels K, Frequin S, Van de Garde EM, Biesma DH, Van der Wees PJ, Van der Nat PB; (SOSMS Consensus Group). Development of an international, multidisciplinary, patient-centered standard outcome set for multiple sclerosis: the S.O.S.MS project. *Multiple Sclerosis and Related Disorders* 2023;**69**:104461. [DOI: [10.1016/j.msard.2022.104461](https://doi.org/10.1016/j.msard.2022.104461)]

67. Carswell A, McColl MA, Baptiste S, Law M, Polatajko H, Pollock N. The Canadian Occupational Performance Measure: a research and clinical literature review. *Canadian Journal of Occupational Therapy* 2004;**71**(4):210-22. [DOI: [10.1177/000841740407100406](https://doi.org/10.1177/000841740407100406)]

68. Law M, Baptiste S, McColl M, Opzoomer A, Polatajko H, Pollock N. The Canadian Occupational Performance Measure: an outcome measure for occupational therapy. *Canadian Journal of Occupational Therapy* 1990;**57**(2):82-7. [DOI: [10.1177/000841749005700207](https://doi.org/10.1177/000841749005700207)]

69. Eyssen IC, Steultjens MP, Oud TA, Bolt EM, Maasdam A, Dekker J. Responsiveness of the Canadian occupational performance measure. *Journal of Rehabilitation Research and Development* 2011;**48**(5):517-28. [DOI: [10.1682/jrrd.2010.06.0110](https://doi.org/10.1682/jrrd.2010.06.0110)]

70. Karhula ME, Kanelisto KJ, Ruutiainen J, Hämäläinen PI, Salminen AL. The activities and participation categories of the ICF Core Sets for multiple sclerosis from the patient perspective. *Disability and Rehabilitation* 2013;**35**(6):492-7. [DOI: [10.3109/09638288.2012.702845](https://doi.org/10.3109/09638288.2012.702845)]

71. Månsson Lexell E, Iwarsson S, Lexell J. The complexity of daily occupations in multiple sclerosis. *Scandinavian Journal of Occupational Therapy* 2006;**13**(4):241-8. [DOI: [10.1080/11038120600840200](https://doi.org/10.1080/11038120600840200)]

- 72.** Pérez de Heredia-Torres M, Huertas-Hoyas E, Sanchez-Camarero C, Maximo-Bocanegra N, Alegre-Ayala J, Sanchez-Herrera-Baeza P, et al. Occupational performance in multiple sclerosis and its relationship with quality of life and fatigue. *European Journal of Physical and Rehabilitation Medicine* 2020;**56**(2):148-54. [DOI: [10.23736/S1973-9087.20.05914-6](https://doi.org/10.23736/S1973-9087.20.05914-6)]
- 73.** Benito-León J, Manuel Morales J, Rivera-Navarro J, Mitchell AJ. A review about the impact of multiple sclerosis on health-related quality of life. *Disability and Rehabilitation* 2003;**25**(23):1291-303. [DOI: [10.1080/09638280310001608591](https://doi.org/10.1080/09638280310001608591)]
- 74.** Hobart J, Lamping D, Fitzpatrick R, Riaz A, Thompson A. The Multiple Sclerosis Impact Scale (MSIS-29): a new patient-based outcome measure. *Brain* 2001;**124**(Pt 5):962-73. [DOI: [10.1093/brain/124.5.962](https://doi.org/10.1093/brain/124.5.962)]
- 75.** Phillips GA, Wyrwich KW, Guo S, Medori R, Altincatal A, Wagner L, et al. Responder definition of the Multiple Sclerosis Impact Scale physical impact subscale for patients with physical worsening. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2014;**20**(13):1753-60. [DOI: [10.1177/1352458514530489](https://doi.org/10.1177/1352458514530489)]
- 76.** Paul L, Coote S, Crosbie J, Dixon D, Hale L, Holloway E, et al. Core outcome measures for exercise studies in people with multiple sclerosis: recommendations from a multidisciplinary consensus meeting. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2014;**20**(12):1641-50. [DOI: [10.1177/1352458514526944](https://doi.org/10.1177/1352458514526944)]
- 77.** Peryer G, Golder S, Junqueira D, Vohra S, Loke YK; on behalf of the Cochrane Adverse Effects Methods Group. Chapter 19: adverse effects. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). In: *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3>. Cochrane, 2022.
- 78.** Cardol M, De Haan RJ, Van den Bos GA, De Jong BA, De Groot IJ. The development of a handicap assessment questionnaire: the Impact on Participation and Autonomy (IPA). *Clinical Rehabilitation* 1999;**13**(5):411-9. [DOI: [10.1191/026921599668601325](https://doi.org/10.1191/026921599668601325)]
- 79.** Cardol M, De Haan RJ, De Jong BA, Van den Bos GA, De Groot IJ. Psychometric properties of the impact on participation and autonomy questionnaire. *Archives of Physical Medicine and Rehabilitation* 2001;**82**(2):210-6. [DOI: [10.1053/apmr.2001.18218](https://doi.org/10.1053/apmr.2001.18218)]
- 80.** Karhula ME, Tolvanen A, Hamalainen PI, Ruutiainen J, Salminen AL, Era P. Predictors of participation and autonomy in people with multiple sclerosis. *American Journal of Occupational Therapy* 2019;**73**(4):7304205070p1-p8. [DOI: [10.5014/ajot.2019.030221](https://doi.org/10.5014/ajot.2019.030221)]
- 81.** Connor KM, Davidson JR. Development of a new resilience scale: the Connor-Davidson Resilience Scale (CD-RISC). *Depression and Anxiety* 2003;**18**(2):76-82. [DOI: [10.1002/da.10113](https://doi.org/10.1002/da.10113)]
- 82.** Broche-Pérez Y, Jiménez-Morales RM, Monasterio-Ramos LO, Bauer J. Validity and reliability of the 10-item Connor-Davidson Resilience Scale (CD-RISC-10) in a sample of Spanish-speaking patients with multiple sclerosis. *Multiple Sclerosis and Related Disorders* 2022;**63**:103914. [DOI: [10.1016/j.msard.2022.103914](https://doi.org/10.1016/j.msard.2022.103914)]
- 83.** Koelmel E, Hughes AJ, Alschuler KN, Ehde DM. Resilience mediates the longitudinal relationships between social support and mental health outcomes in multiple sclerosis. *Archives of Physical Medicine and Rehabilitation* 2017;**98**(6):1139-48. [DOI: [10.1016/j.apmr.2016.09.127](https://doi.org/10.1016/j.apmr.2016.09.127)]
- 84.** Swanepoel I, Van Staden W, Fletcher L. Psychological resilience and vulnerability as mediators between adverse life events and fatigue, motor dysfunction, and paresthesia in multiple sclerosis. *Psychosomatic Medicine* 2020;**82**(2):138-46. [DOI: [10.1097/PSY.0000000000000770](https://doi.org/10.1097/PSY.0000000000000770)]
- 85.** Chiu CY, Motl RW. Further validation of the multiple sclerosis self-efficacy scale. *Disability and Rehabilitation* 2015;**37**(26):2429-38. [DOI: [10.3109/09638288.2015.1021022](https://doi.org/10.3109/09638288.2015.1021022)]
- 86.** Schwartz CE, Coulthard-Morris L, Zeng Q, Retzlaff P. Measuring self-efficacy in people with multiple sclerosis: a validation study. *Archives of Physical Medicine and Rehabilitation* 1996;**77**(4):394-8. [DOI: [10.1016/s0003-9993\(96\)90091-x](https://doi.org/10.1016/s0003-9993(96)90091-x)]
- 87.** Kayes NM, McPherson KM, Schluter P, Taylor D, Leete M, Kolt GS. Exploring the facilitators and barriers to engagement in physical activity for people with multiple sclerosis. *Disability and Rehabilitation* 2011;**33**(12):1043-53. [DOI: [10.3109/09638288.2010.520801](https://doi.org/10.3109/09638288.2010.520801)]
- 88.** Sinnakaruppan I, Macdonald K, McCafferty A, Mattison P. An exploration of the relationship between perception of control, physical disability, optimism, self-efficacy and hopelessness in multiple sclerosis. *International Journal of Rehabilitation Research* 2010;**33**(1):26-33. [DOI: [10.1097/MRR.0b013e32832e6b16](https://doi.org/10.1097/MRR.0b013e32832e6b16)]
- 89.** Bishop M, Frain MP. Development and initial analysis of multiple sclerosis self-management scale. *International Journal of MS Care* 2007;**9**(2):35-42. [DOI: [10.7224/1537-2073.2018-090](https://doi.org/10.7224/1537-2073.2018-090)]
- 90.** Bishop M, Frain MP. The multiple sclerosis self-management scale: revision and psychometric analysis. *Rehabilitation Psychology* 2011;**56**(2):150-9. [DOI: [10.1037/a0023679](https://doi.org/10.1037/a0023679)]
- 91.** Efendi H, Ünal A, Akçalı A, Altunhan B, Bingöl A, Altunrende B, et al. The effect of cognitive performance on self-management behavior of multiple sclerosis patients. *Multiple Sclerosis and Related Disorders* 2022;**63**:103880. [DOI: [10.1016/j.msard.2022.103880](https://doi.org/10.1016/j.msard.2022.103880)]
- 92.** Wilski M, Tasiemski T. Illness perception, treatment beliefs, self-esteem, and self-efficacy as correlates of self-management in multiple sclerosis. *Acta Neurologica Scandinavica* 2016;**133**(5):338-45. [DOI: [10.1111/ane.12465](https://doi.org/10.1111/ane.12465)]
- 93.** Bjelland I, Dahl AA, Haug TT, Neckelmann D. The validity of the Hospital Anxiety and Depression Scale. an updated literature review. *Journal of Psychosomatic Research* 2002;**52**(2):69-77. [DOI: [10.1016/s0022-3999\(01\)00296-3](https://doi.org/10.1016/s0022-3999(01)00296-3)]

94. Zigmond AS, Snaith RP. The Hospital Anxiety and Depression Scale. *Acta Psychiatrica Scandinavica* 1983;**67**(6):361-70. [DOI: [10.1111/j.1600-0447.1983.tb09716.x](https://doi.org/10.1111/j.1600-0447.1983.tb09716.x)]
95. Allataifeh E, Khalil H, Almhdawi K, Al-Shorman A, Hawamdeh Z, El-Salem K et al. The clinical correlates of participation levels in people with multiple sclerosis. *NeuroRehabilitation* 2020;**47**(2):153-60. [DOI: [10.3233/NRE-203131](https://doi.org/10.3233/NRE-203131)]
96. Wu Y, Levis B, Sun Y, He C, Krishnan A, Neupane D, et al. DEPRESSION Screening Data (DEPRESSD) HADS Group. Accuracy of the Hospital Anxiety and Depression Scale Depression subscale (HADS-D) to screen for major depression: systematic review and individual participant data meta-analysis. *BMJ (Clinical Research Ed.)* 2021;**373**:n972. Erratum in: *The BMJ*. 2021 May 19;373:n1231. [DOI: [10.1136/bmj.n972](https://doi.org/10.1136/bmj.n972)]
97. Robinson BC. Validation of a caregiver strain index. *Journal of Gerontology* 1983;**38**(3):344-8. [DOI: [10.1093/geronj/38.3.344](https://doi.org/10.1093/geronj/38.3.344)]
98. Thornton M, Travis SS. Analysis of the reliability of the modified caregiver strain index. *Journals of Gerontology - Psychological Sciences and Social Sciences* 2003;**58**(2):S127-32. [DOI: [10.1093/geronb/58.2.s127](https://doi.org/10.1093/geronb/58.2.s127)]
99. García-Domínguez JM, Martínez-Ginés ML, Carmona O, Caminero AB, Prefasi D, Maurino J et al. W-IMPACT Clinical Investigators. Measuring burden in caregivers of people with multiple sclerosis: psychometric properties of the CSI questionnaire. *Patient Preference and Adherence* 2019;**13**:101-6. [DOI: [10.2147/PPA.S180863](https://doi.org/10.2147/PPA.S180863)]
100. Sarhan AA, El-Sharkawy KA, Mahmoudy AM, Hashim NA. Burden of multiple sclerosis: impact on the patient, family and society. *Multiple Sclerosis and Related Disorders* 2022;**63**:103864. [DOI: [10.1016/j.msard.2022](https://doi.org/10.1016/j.msard.2022)]
101. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. Updating guidance for reporting systematic reviews: development of the PRISMA 2020 statement. *Journal of Clinical Epidemiology* 2021;**134**:103-12. [DOI: [10.1016/j.jclinepi.2021.02.003](https://doi.org/10.1016/j.jclinepi.2021.02.003)]
102. Lefebvre C, Glanville J, Briscoe S, Littlewood A, Marshall C, Metzendorf M-I, et al. Chapter 4: Searching for and selecting studies. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). In: *Cochrane Handbook for Systematic Reviews of Interventions* version 6.3 (updated February 2022). Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3>. Cochrane, 2022.
103. Rethlefsen ML, Kirtley S, Waffenschmidt S, Ayala AP, Moher D, Page MJ, et al; Prisma-S Group. PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. *Systematic Reviews* 2021;**10**(1):39. [DOI: [10.1186/s13643-020-01542-z](https://doi.org/10.1186/s13643-020-01542-z)]
104. Avau B, Van Remoortel H, De Buck E. Translation and validation of PubMed and Embase search filters for identification of systematic reviews, intervention studies, and observational studies in the field of first aid. *Journal of the Medical Library Association* 2021;**109**(4):599-608. [DOI: [10.5195/jmla.2021.1219](https://doi.org/10.5195/jmla.2021.1219)]
105. Salomé A, Sasso D'Elia T, Franchini G, Santilli V, Paolucci T. Occupational therapy in fatigue management in multiple sclerosis: an umbrella review. *Multiple Sclerosis International* 2019;**March 21**:1-7. [DOI: [10.1155/2019/2027947](https://doi.org/10.1155/2019/2027947)]
106. Belveal K, Gunkel-Lam S, Hajare A, Lambropoulos A, Rogers S, Hilton C, et al. The effectiveness of nontraditional or home-based programming on ADL performance of individuals living with multiple sclerosis: a systematic review. *Multiple Sclerosis and Related Disorders* 2023;**71**:104576. [DOI: [10.1016/j.msard.2023.104576](https://doi.org/10.1016/j.msard.2023.104576)]
107. EndNote. The EndNote Team, Version EndNote X9. Philadelphia, PA: Clarivate, 2013.
108. Review Manager (RevMan). Version 8.18.0. The Cochrane Collaboration, 2025. Available at <https://revman.cochrane.org>.
109. Hoffmann TC, Glasziou PP, Boutron I, Milne R, Perera R, Moher D, et al. Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide. *BMJ (Clinical Research Ed)* 2014;**348**:g1687. [DOI: [10.1136/bmj.g1687](https://doi.org/10.1136/bmj.g1687)]
110. Sterne JA, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ (Clinical Research Ed.)* 2019;**366**:l4898. [DOI: [10.1136/bmj.l4898](https://doi.org/10.1136/bmj.l4898)]
111. Higgins JP, Savović J, Page MJ, Elbers RG, Sterne JA. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al. *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.3 (updated February 2022). Cochrane, 2022. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3>.
112. Beckerman H, Eijssen I, Walraven R, Strijbis EM, Terwee CB, Uitdehaag BM, et al. Which type of PROMs to use in MS routine clinical care: the validity of PROMIS CAT questionnaires. *Multiple Sclerosis and Related Disorders* 2025;**95**:106320. [DOI: [10.1016/j.msard.2025.106320](https://doi.org/10.1016/j.msard.2025.106320)]
113. Ciani O, Manyara AM, Davies P, Stewart D, Weir CJ, Young AE, et al. A framework for the definition and interpretation of the use of surrogate endpoints in interventional trials. *EClinicalMedicine* 2023;**65**:102283. [DOI: [10.1016/j.eclinm.2023.102283](https://doi.org/10.1016/j.eclinm.2023.102283)]
114. Knoop H, Van Kessel K, Moss-Morris R. Which cognitions and behaviours mediate the positive effect of cognitive behavioural therapy on fatigue in patients with multiple sclerosis? *Psychological Medicine* 2012;**42**(1):205-13. [DOI: [10.1017/S0033291711000924](https://doi.org/10.1017/S0033291711000924)]
115. Sterne JA, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, et al. ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *BMJ (Clinical Research Ed.)* 2016;**355**:i4919. [DOI: [10.1136/bmj.i4919](https://doi.org/10.1136/bmj.i4919)]

- 116.** Karim ME, Tremlett H, Zhu F, Petkau J, Kingwell E. Dealing with treatment-confounder feedback and sparse follow-up in longitudinal studies: application of a marginal structural model in a multiple sclerosis cohort. *American Journal of Epidemiology* 2021;**190**(5):908-17. [DOI: [10.1093/aje/kwaa243](https://doi.org/10.1093/aje/kwaa243)]
- 117.** Marrie RA. Comorbidity in multiple sclerosis: implications for patient care. *Nature Reviews Neurology* 2017;**13**(6):375-82. [DOI: [10.1038/nrneuro.2017.33](https://doi.org/10.1038/nrneuro.2017.33)]
- 118.** Suero M, Botella J, Durán JI. Methods for estimating the sampling variance of the standardized mean difference. *Psychological Methods* 2021;**Dec 16**:Online ahead of print. [DOI: [10.1037/met0000446](https://doi.org/10.1037/met0000446)]
- 119.** Gallardo-Gómez D, Richardson R, Dwan K. Standardized mean differences in meta-analysis: a tutorial. *Cochrane Evidence Synthesis and Methods* 2024;**2**:e12047. [DOI: [10.1002/cesm.12047](https://doi.org/10.1002/cesm.12047)]
- 120.** Jakobsen JC, Gluud C, Wetterslev J, Winkel P. When and how should multiple imputation be used for handling missing data in randomised clinical trials - a practical guide with flowcharts. *BMC Medical Research Methodology* 2017;**17**(1):162. [DOI: [10.1186/s12874-017-0442-1](https://doi.org/10.1186/s12874-017-0442-1)]
- 121.** Furukawa TA, Barbui C, Cipriani A, Brambilla P, Watanabe N. Imputing missing standard deviations in meta-analyses can provide accurate results. *Journal of Clinical Epidemiology* 2006;**59**(1):7-10. [DOI: [10.1016/j.jclinepi.2005.06.006](https://doi.org/10.1016/j.jclinepi.2005.06.006)]
- 122.** Higgins JP, Li T, Deeks JJ (editors). Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.4>.
- 123.** Peters JL, Sutton AJ, Jones DR, Abrams KR, Rushton L. Contour-enhanced meta-analysis funnel plots help distinguish publication bias from other causes of asymmetry. *Journal of Clinical Epidemiology* 2008;**61**(10):991-6. [DOI: [10.1016/j.jclinepi.2007.11.010](https://doi.org/10.1016/j.jclinepi.2007.11.010)]
- 124.** Deeks JJ, Higgins JP, Altman DG, McKenzie JE, Veroniki AA (editors). Chapter 10: Analysing data and undertaking meta-analyses (last updated November 2024). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s).. *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://cochrane.org/handbook>.
- 125.** Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *BMJ (Clinical Research Ed.)* 2020;**368**:l6890. [DOI: [10.1136/bmj.l6890](https://doi.org/10.1136/bmj.l6890)]
- 126.** Schünemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions (last updated August 2023). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). In: *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Available from <https://www.cochrane.org/handbook>.
- 127.** Cohen J. Statistical Power Analysis for the Behavioral Sciences. 2nd edition. UK: Taylor & Francis, 2013. [ISBN: 9780203771587]
- 128.** Hammel KW. Building globally relevant occupational therapy from the strength of our diversity. *World Federation of Occupational Therapists Bulletin* 2019;**75**(1):13-26. [DOI: [10.1080/14473828.2018.1529480](https://doi.org/10.1080/14473828.2018.1529480)]
- 129.** MSIF (The Multiple Sclerosis International Federation). Atlas of MS 3rd Edition: PART 2: Clinical management of multiple sclerosis around the world: Key findings about the diagnosis and clinical management of MS. MSIF April 2021 (accessed June 2022). [MSIF: www.msif.org/wp-content/uploads/2021/05/Atlas-3rd-Edition-clinical-management-report-EN-5-5-21.pdf]
- 130.** Roddam H, Rog D, Janssen J, Wilson N, Cross L, Olajide O, et al. Inequalities in access to health and social care among adults with multiple sclerosis: a scoping review of the literature. *Multiple Sclerosis and Related Disorders* 2019;**28**:290-304. [DOI: [10.1016/j.msard.2018.12.043](https://doi.org/10.1016/j.msard.2018.12.043)]
- 131.** World Federation of Occupational Therapists (WFOT). Human Resources Project 2018: Global demographics of the occupational therapy profession. www.wfot.org/resources/2018-occupational-therapy-human-resources-project-full-alphabetical 2018 (accessed June 2022).
- 132.** Welch VA, Petkovic J, Jull J, Hartling L, Klassen T, Kristjansson E, et al. Chapter 16: Equity and specific populations (last updated October 2019). In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.5 (updated August 2024). Cochrane, 2024. Available from <https://cochrane.org/handbook>.
- 133.** O'Neill J, Tabish H, Welch V, Petticrew M, Pottie K, Clarke M, et al. Applying an equity lens to interventions: Using PROGRESS ensures consideration of socially stratifying factors to illuminate inequities in health. *Journal of Clinical Epidemiology* 2014;**67**(1):56-64. [DOI: [10.1016/j.jclinepi.2013.08.005](https://doi.org/10.1016/j.jclinepi.2013.08.005)]
- 134.** Ueffing E, Tugwell P, Welch V, Petticrew M, Kristjansson E for the Campbell and Cochrane Equity Methods Group. Equity checklist for systematic review authors version 2012-10-0. <https://methods.cochrane.org/sites/methods.cochrane.org/equity/files/uploads/EquityChecklist2012.pdf> (accessed June 2022) 2011.
- 135.** Atkins D, Best D, Briss PA, Eccles M, Falck-Ytter Y, Flottorp S et al. Grading quality of evidence and strength of recommendations. *BMJ (Clinical Research Ed.)* 2004;**328**(7454):1490. [DOI: [10.1136/bmj.328.7454.1490](https://doi.org/10.1136/bmj.328.7454.1490)]
- 136.** Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ (Clinical Research Ed.)* 2008;**336**(7650):924-6. [DOI: [10.1136/bmj.39489.470347.AD](https://doi.org/10.1136/bmj.39489.470347.AD)]

- 137.** Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2011;**64**(4):383-94. [DOI: [10.1016/j.jclinepi.2010.04.026](https://doi.org/10.1016/j.jclinepi.2010.04.026)]
- 138.** GRADEpro GDT. Version accessed May 2nd 2025. Hamilton (ON): McMaster University (developed by Evidence Prime), 2021. Available at <https://grade.pro.org>.
- 139.** Schünemann HJ, Higgins JP, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). In: Cochrane Handbook for Systematic Reviews of Interventions Version 6.3 (updated February 2022). Available from <https://www.cochrane.org/authors/handbooks-and-manuals/handbook/archive/v6.3>. Cochrane, 2022.
- 140.** Santesso N, Carrasco-Labra A, Langendam M, Brignardello-Petersen R, Mustafa RA, Heus P, et al. Improving GRADE evidence tables part 3: detailed guidance for explanatory footnotes supports creating and understanding GRADE certainty in the evidence judgments. *Journal of Clinical Epidemiology* 2016;**74**:28-39. [DOI: [10.1016/j.jclinepi.2015.12.006](https://doi.org/10.1016/j.jclinepi.2015.12.006)]
- 141.** Akbarfahimi M, Nabavi SM, Kor B, Rezaie L, Paschall E. The effectiveness of occupational therapy-based sleep interventions on quality of life and fatigue in patients with multiple sclerosis: a pilot randomized clinical trial study. *Neuropsychiatric Disease and Treatment* 2020;**16**:1369-79. [DOI: [10.2147/NDT.S249277](https://doi.org/10.2147/NDT.S249277)]
- 142.** Eyssen IC, Steultjens MP, De Groot V, Steultjens EM, Knol DL, Polman CH, et al. A cluster randomised controlled trial on the efficacy of client-centred occupational therapy in multiple sclerosis: good process, poor outcome. *Disability and Rehabilitation* 2013;**35**(19):1636-46. [DOI: [10.3109/09638288.2012.748845](https://doi.org/10.3109/09638288.2012.748845)]
- 143.** Finlayson M, Preissner K, Cho C, Plow M. Randomized trial of a teleconference-delivered fatigue management program for people with multiple sclerosis. *Multiple Sclerosis Journal* 2011;**17**(9):1130-40. [DOI: [10.1177/1352458511404272](https://doi.org/10.1177/1352458511404272)]
- 144.** Garcia-Burguillo Ma DP, Aguila-Maturana AM. Energy-saving strategies in the treatment of fatigue in patients with multiple sclerosis. A pilot study. *Revista de Neurologia* 2009;**49**(4):181-5. [DOI: [10.33588/rn.4904.2008379](https://doi.org/10.33588/rn.4904.2008379)]
- 145.** Hersche R, Weise A, Della Bella S, Michel G, Marco M, Kool J. Feasibility and preliminary results of a short inpatient energy-management education for person with MS-fatigue. *Archives of Physical Medicine and Rehabilitation* 2019;**100**(10):e77. [DOI: [10.1016/j.apmr.2019.08.223](https://doi.org/10.1016/j.apmr.2019.08.223)]
- 146.** Hynes SM, Dwyer CP, Alvarez-Iglesias A, Rogers F, Joyce RA, Oglesby MH, et al; Charamba B; COB-MS PPI Advisory Panel. A cluster-randomised controlled feasibility trial evaluating the Cognitive Occupation-Based programme for people with Multiple Sclerosis (COB-MS). *Neurological Sciences* 2024;**46**(1):445-62. [DOI: [10.1007/s10072-024-07757-5](https://doi.org/10.1007/s10072-024-07757-5)]
- 147.** Kos D, Duportail M, Meirte J, Meeus M, D'Hooghe M, Nagels, et al. The effectiveness of a self-management occupational therapy intervention on activity performance in individuals with multiple sclerosis-related fatigue: a randomized-controlled trial. *International Journal of Rehabilitation Research. Internationale Zeitschrift für Rehabilitationsforschung. Revue Internationale de Recherches de Readaptation* 2016;**39**(3):255-62. [DOI: [10.1097/MRR.0000000000000178](https://doi.org/10.1097/MRR.0000000000000178)]
- 148.** Kalina JT, Hinojosa J, Strober L, Bacon J, Donnelly S, Goverover Y. Randomized controlled trial to improve self-efficacy in people with multiple sclerosis: the community reintegration for socially isolated patients (CRISP) program. *American Journal of Occupational Therapy: Official Publication of the American Occupational Therapy Association* 2018;**72**(5):7205205030p1-7205205030p8. [DOI: [10.5014/ajot.2018.026864](https://doi.org/10.5014/ajot.2018.026864)]
- 149.** Blikman LJ, Van Meeteren J, Twisk JW, De Laat FA, De Groot V, Beckerman H, et al. Effectiveness of energy conservation management on fatigue and participation in multiple sclerosis: a randomized controlled trial. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2017;**23**(11):1527-41. [DOI: [10.1177/1352458517702751](https://doi.org/10.1177/1352458517702751)]
- 150.** Mathiowetz VG, Finlayson ML, Matuska KM, Chen HY, Luo P. Randomized controlled trial of an energy conservation course for persons with multiple sclerosis. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2005;**11**(5):592-601. [DOI: [10.1191/1352458505ms1198oa](https://doi.org/10.1191/1352458505ms1198oa)]
- 151.** Plow M, Finlayson M, Liu J, Motl RW, Bethoux F, Sattar A. Randomized controlled trial of a telephone-delivered physical activity and fatigue self-management interventions in adults with multiple sclerosis. *Archives of Physical Medicine and Rehabilitation* 2019;**100**(11):2006-14. [DOI: [10.1016/j.apmr.2019.04.022](https://doi.org/10.1016/j.apmr.2019.04.022)]
- 152.** Vanage SM, Gilbertson KK, Mathiowetz V. Effects of an energy conservation course on fatigue impact for persons with progressive multiple sclerosis. *American Journal of Occupational Therapy*. 2003;**57**(3):315-23. [DOI: [10.5014/ajot.57.3.315](https://doi.org/10.5014/ajot.57.3.315)]
- 153.** Thomas S, Thomas PW, Kersten P, Jones R, Green C, Nock A, et al. A pragmatic parallel arm multi-centre randomised controlled trial to assess the effectiveness and cost-effectiveness of a group-based fatigue management programme (FACETS) for people with multiple sclerosis. *Journal of Neurology, Neurosurgery and Psychiatry* 2013;**84**(10):1092-9. [DOI: [10.1136/jnnp-2012-303816](https://doi.org/10.1136/jnnp-2012-303816)]
- 154.** Cuesta-Gomez A, Martin-Diaz P, Baeza PS, Martinez-Medina A, Ortiz-Comino C, Cano-de-la-Cuerda R. Nintendo switch joy-cons' infrared motion camera sensor for training manual dexterity in people with multiple sclerosis: a randomized controlled trial. *Journal of Clinical Medicine* 2022;**11**(12):12. [DOI: [10.3390/jcm11123261](https://doi.org/10.3390/jcm11123261)]
- 155.** Goverover Y, Chiaravalloti N, Genova H, DeLuca J. A randomized controlled trial to treat impaired learning and memory in multiple sclerosis: the self-GEN trial. *Multiple Sclerosis Journal* 2018;**24**(8):1096-104. [DOI: [10.1177/1352458517709955](https://doi.org/10.1177/1352458517709955)]

- 156.** Jones L, Lewis Y, Harrison J, Wiles CM. The effectiveness of occupational therapy and physiotherapy in multiple sclerosis patients with ataxia of the upper limb and trunk. *Clinical Rehabilitation* 1996;**10**(4):277-82. [DOI: [10.1177/026921559601000403](https://doi.org/10.1177/026921559601000403)]
- 157.** Sanchez-Herrera-Baeza P, Garcia-Bravo C, Huertas-Hoyas E, Florencio LL, Martinez-Piedrola RM, Pérez-Corrales J, et al. Mental practice and manipulative skills training among people with multiple sclerosis: a pilot study. *American Journal of Occupational Therapy*. 2022;**76**(2):7602205040. [DOI: [10.5014/ajot.2022.044479](https://doi.org/10.5014/ajot.2022.044479)]
- 158.** Askari S, Kessler D, Smyth P, Finlayson M. Evaluating occupational performance coaching to support fatigue management for people with multiple sclerosis: a feasibility study. *Clinical Rehabilitation* 2022;**36**(9):1244-56. [DOI: [10.1177/02692155221107074](https://doi.org/10.1177/02692155221107074)]
- 159.** Kessler D, Franz M, Malakouti N, Rajachandrakumar R, Baharnoori M, Finlayson M. Randomized controlled trial of occupational performance coaching for adults with multiple sclerosis. *Archives of Physical Medicine and Rehabilitation* 2024;**105**(9):1649-56. [DOI: [10.1016/j.apmr.2024.05.022](https://doi.org/10.1016/j.apmr.2024.05.022)]
- 160.** Kamm CP, Mattle HP, Müri RM, Heldner MR, Blatter V, Bartlome S, et al. Home-based training to improve manual dexterity in patients with multiple sclerosis: a randomized controlled trial. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2015;**21**(12):1546-56. [DOI: [10.1177/1352458514565959](https://doi.org/10.1177/1352458514565959)]
- 161.** Thomas PW, Thomas S, Kersten P, Jones R, Slingsby V, Nock A, et al. One year follow-up of a pragmatic multi-centre randomised controlled trial of a group-based fatigue management programme (FACETS) for people with multiple sclerosis. *BMC Neurology* 2014;**14**:109. [DOI: [10.1186/1471-2377-14-109](https://doi.org/10.1186/1471-2377-14-109)]
- 162.** Sterne J, Higgins J. ROBINS-I V2 tool. Cochrane Methods Bias: Risk of bias tools. Available at <https://www.riskofbias.info/welcome/robins-i-v2> 2024.
- 163.** Das Nair R, De Groot V, Freeman J. Beyond current research practice: methodological considerations in MS rehabilitation research (is designing the perfect rehabilitation trial the Holy Grail or a Gordian knot?). *Multiple Sclerosis Journal* 2019;**25**(10):1337-47. [DOI: [10.1177/1352458519858271](https://doi.org/10.1177/1352458519858271)]
- 164.** Hayward KS, Dalton EJ, Barth J, Brady M, Cherney LR, Churilov L, et al. Control intervention design for preclinical and clinical trials: consensus-based core recommendations from the third Stroke Recovery and Rehabilitation Roundtable. *Neurorehabilitation and Neural Repair* 2024;**38**(1):30-40. [DOI: [10.1177/15459683231209162](https://doi.org/10.1177/15459683231209162)]
- 165.** Motl RW. Randomized controlled trials of rehabilitation in multiple sclerosis: barriers and unmet needs. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2024;**30**(5_suppl):36-40. [DOI: [10.1177/13524585241292950](https://doi.org/10.1177/13524585241292950)]
- 166.** Soler B, Raats J, Abasiyanik Z, Lamers I, Makshakov G, Feys P. Systematic evaluation of the guidelines for rehabilitation in multiple sclerosis patients: an overview according to ICF functioning domains. *International Journal of Rehabilitation Research* 2021;**44**(4):289-97. [DOI: [10.1097/mrr.0000000000000501](https://doi.org/10.1097/mrr.0000000000000501)]
- 167.** Yost E, Trojanowski S, Jarnigin T, De Bear K, Yorke A. Examining occupation-based outcome measure use in a neurologic outpatient occupational therapy setting: a knowledge translation initiative. *Translational Science in Occupation* 2024;**1**(3):1. [DOI: [10.32873/unmc.dc.tso.1.3.01](https://doi.org/10.32873/unmc.dc.tso.1.3.01)]
- 168.** Finlayson M, Feys P, Dalgas U, Kos D. Intermediate outcomes for clinical trials of multiple sclerosis rehabilitation interventions: Conceptual and practical considerations. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2023;**29**(9):1186-94. [DOI: [10.1177/13524585231189674](https://doi.org/10.1177/13524585231189674)]
- 169.** Cella D, Yount S, Rothrock N, Gershon R, Cook K, Reeve B, et al. PROMIS cooperative group. The patient-reported outcomes measurement information system (PROMIS): progress of an NIH roadmap cooperative group during its first two years. *Medical Care* 2007;**45**(5 Suppl 1):S3-S11. [DOI: [10.1097/01.mlr.0000258615.42478.55](https://doi.org/10.1097/01.mlr.0000258615.42478.55)]
- 170.** Benson T. Why proms and prems matter? In: Patient-Reported Outcomes and Experience: Measuring What We Want from PROMs and PREMs. Cham: Springer International Publishing, 2022:3-12. [DOI: [10.1007/978-3-030-97071-0_1](https://doi.org/10.1007/978-3-030-97071-0_1)]
- 171.** Larsen AE, Law M. Canadian Occupational Performance Measure the state of the art - a review. *Scandinavian Journal of Occupational Therapy* 2025;**32**(1):2473045. [DOI: [10.1080/11038128.2025.2473045](https://doi.org/10.1080/11038128.2025.2473045)]
- 172.** Rajachandrakumar R, Finlayson M. Multiple sclerosis caregiving: a systematic scoping review to map current state of knowledge. *Health & Social Care in the Community* 2022;**30**(4):e874-e897. [DOI: [10.1111/hsc.13687](https://doi.org/10.1111/hsc.13687)]
- 173.** Adal TG, Van der Mei I, Taylor BV, De Graaff B, Palmer AJ, Chen G, et al. Investigation of the health economic analysis of informal care for people living with a chronic neurological disease: a systematic review and meta-analysis of the global evidence for multiple sclerosis. *Social Science & Medicine* 2024;**363**:117405. [DOI: [10.1016/j.socscimed.2024.117405](https://doi.org/10.1016/j.socscimed.2024.117405)]
- 174.** Feys P, Moghames P, Kasilingam E, King R, Manacorda T, Fonseca M, et al. Prevalence, severity and impact of MS symptoms, and use of services in Europe: results from the large scale IMSS survey. *Multiple Sclerosis Journal* 2024;**30**(3S):1148-211. [DOI: [10.1177/13524585241269220](https://doi.org/10.1177/13524585241269220)]
- 175.** Bebo BF Jr, Banwell BL, Whitacre CC, Coetzee T, Dalgas U, De Jager PL, et al. The refined Pathways to Cures Research Roadmap for multiple sclerosis cures. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2024;**30**(10):1242-51. [DOI: [10.1177/13524585241266483](https://doi.org/10.1177/13524585241266483)]
- 176.** Wong G. Realist reviews in health policy and systems research. In: Langlois ÉV, Daniels K, Akl EA, editor(s). Evidence Synthesis for Health Policy and Systems: a Methods Guide. Geneva: World Health Organization, 2018.
- 177.** Dover DC, Belon AP. The health equity measurement framework: a comprehensive model to measure social

inequities in health. *International Journal for Equity in Health* 2019;**18**(1):1-12. [DOI: [10.1186/s12939-019-0935-0](https://doi.org/10.1186/s12939-019-0935-0)]

178. World Health Organization. A conceptual framework for action on the social determinants of health. <https://apps.who.int/iris/handle/10665/44489> 2010 (accessed June 2022).

179. STEP UP. Strategies for trialists to promote equal participation in clinical trials for under-served populations. <https://step-up-clinical-trials.co.uk/> (accessed 3 June 2024).

180. Treweek S, Banister K, Bower P, Cotton S, Devane D, Gardner HR, et al. Developing the INCLUDE Ethnicity Framework-a tool to help trialists design trials that better reflect the communities they serve. *Trials* 2021;**22**(1):337. [DOI: [10.1186/s13063-021-05276-8](https://doi.org/10.1186/s13063-021-05276-8)]

181. Joyce R, Dwyer CP, Hynes SM. Twelve months into a feasibility trial: reflections on three experiences of public and patient involvement in research. *HRB Open Research* 2021;**4**:11. [DOI: [10.12688/hrbopenres.13205.2](https://doi.org/10.12688/hrbopenres.13205.2)]

182. Soelberg Sorensen P, Giovannoni G, Montalban X, Thalheim C, Zaratin P, Comi G. The multiple sclerosis care unit. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2019;**25**(5):627-36. [DOI: [10.1177/1352458518807082](https://doi.org/10.1177/1352458518807082)]

183. Savio M, Kearney H, Giunti G. Evaluating the evidence behind multidisciplinary roles for a multiple sclerosis unit: a systematic literature review. *Multiple Sclerosis and Related Disorders* 2025;**95**:106342. [DOI: [10.1016/j.msard.2025.106342](https://doi.org/10.1016/j.msard.2025.106342)]

184. Harrison AM, Safari R, Mercer T, Picariello F, Van der Linden ML, White C, et al. Which exercise and behavioural interventions show most promise for treating fatigue in multiple sclerosis? A network meta-analysis. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2021;**27**(11):1657-167. [DOI: [10.1177/1352458521996002](https://doi.org/10.1177/1352458521996002)]

185. Junqueira DR, Zorzela L, Golder S, Loke Y, Gagnier JJ, Julious SA, et al; CONSORT Harms Group (2023). CONSORT Harms 2022 statement, explanation, and elaboration: updated guideline for the reporting of harms in randomised trials. *BMJ (Clinical Research Ed.)* 2023;**381**:e073725.

186. Khan F, Amatya B, Kesselring J, Galea M. Telerehabilitation for persons with multiple sclerosis. *Cochrane Database of*

Systematic Reviews 2015, Issue 4. Art. No: CD010508. [DOI: [10.1002/14651858.CD010508.pub2](https://doi.org/10.1002/14651858.CD010508.pub2)]

187. Prosperini L, Arrambide G, Celius EG, Goletti D, Killestein J, Kos D, et al. COVID-19 and multiple sclerosis: challenges and lessons for patient care. *Lancet Regional Health-Europe* 2024;**44**:100979. [DOI: [10.1016/j.lanepe.2024.100979](https://doi.org/10.1016/j.lanepe.2024.100979)]

188. Alwadai B, Lazem H, Almoajil H, Hall AJ, Mansoubi M, Dawes H. Telerehabilitation and its impact following stroke: an umbrella review of systematic reviews. *Journal of Clinical Medicine* 2024;**14**(1):50. [DOI: [10.3390/jcm14010050](https://doi.org/10.3390/jcm14010050)]

189. Khan F, Turner-Stokes L, Ng L, Kilpatrick T, Amatya B. Multidisciplinary rehabilitation for adults with multiple sclerosis. *Cochrane Database of Systematic Reviews* 2007, Issue 2. Art. No: CD006036. [DOI: [10.1002/14651858.CD006036.pub2](https://doi.org/10.1002/14651858.CD006036.pub2)]

190. World Health Organization. International Classification of Health Interventions (ICHI). www.who.int/standards/classifications/international-classification-of-health-interventions (accessed January 2026).

191. Hopewell S, Chan AW, Collins GS, Hróbjartsson A, Moher D, Schulz KF, et al. CONSORT 2025 statement: updated guideline for reporting randomised trials. *BMJ (Clinical Research Ed)* 2025;**389**:e081123. [DOI: [10.1136/bmj-2024-081123](https://doi.org/10.1136/bmj-2024-081123)]

192. National Institute for Health Research (NIHR). The INCLUDE Ethnicity Framework. <https://www.trialforge.org/trial-diversity/include/> (accessed January 2026).

193. IRCT20120910010806N6. Quality of sleep, life and fatigue in patient with multiple sclerosis. <https://trialsearch.who.int/Trial2.aspx?TrialID=IRCT20120910010806N6> (first registered 10 March 2018).

194. Hynes SM, Forwell S. A cognitive occupation-based programme for people with multiple sclerosis: a new occupational therapy cognitive rehabilitation intervention. *Hong Kong Journal of Occupational Therapy* 2019;**32**(1):41-52. [DOI: [10.1177/1569186119841263](https://doi.org/10.1177/1569186119841263)]

195. Plow M, Motl RW, Finlayson M, Bethoux F. Intervention mediators in a randomized controlled trial to increase physical activity and fatigue self-management behaviors among adults with multiple sclerosis. *Annals of Behavioral Medicine* 2020;**54**(3):213-21. [DOI: [10.1093/abm/kaz033](https://doi.org/10.1093/abm/kaz033)]

ADDITIONAL TABLES

Table 1. Overview of included studies, outcomes and synthesis

Study Year Country	Study type	Intervention	Control	Sample size (intervention/ control)	Outcome with avail- able data	Specific outcome measure	Time points of mea- surement*	Synthesis method
Akbarfahimi 2020 [141, 193] Iran	RCT	Fatigue manage- ment	Usual care	20 (10/10)	Daily func- tioning	Fatigue Impact Scale (FIS)	Post-intervention	MA
					Quality of life	Short Form 36 (SF-36)	Post-intervention	MA
Askari 2022 [158] Canada	RCT	Occupa- tional per- formance coaching (telephone)	MS INFoRm website	24 (12/12)	Daily func- tioning	Modified Fatigue Impact Scale (MFIS)	Post-intervention	MA
					Self-efficacy	Multiple Sclerosis Self-Efficacy Scale (MSSES)	Post-intervention	MA
Blikman 2017 [149] The Nether- lands	RCT	Fatigue manage- ment	Information only	84 (41/43)	Daily func- tioning	Modified Fatigue Impact Scale (MFIS)	Post-intervention Medium-term fol- low-up Long-term follow-up	MA
					Quality of life	Short Form 36 (SF-36)	Post-intervention Medium-term fol- low-up Long-term follow-up	MA
					Participation	Impact on Participation and Au- tonomy scale (IPA)	Post-intervention Medium-term fol- low-up Long-term follow-up	MA

Table 1. Overview of included studies, outcomes and synthesis (Continued)

Cues- ta-Gomez 2022 [154] Spain	RCT	Brain Train- ing Video Game upper limb train- ing	Convention- al upper limb train- ing	21 (11/10)	Quality of life	Multiple Sclerosis Impact Scale (MSIS-29)	Post-intervention	MA
Eyssen 2013 [142] The Nether- lands	RCT	Client-cen- tred occu- pational therapy	Usual oc- cupational therapy	269 (156/113)	Daily func- tioning	Canadian Occupational Perfor- mance Measure (COPM)	Post-intervention Medium-term fol- low-up	NS
					Quality of life	Short Form 36 (SF-36)	Post-intervention Medium-term fol- low-up	NS
Finlayson 2011 [143] USA	RCT	Fatigue manage- ment (tele- conference)	Waiting list	181 (89/92)	Daily func- tioning	Fatigue Impact Scale (FIS)	Post-intervention	MA
					Quality of life	Short Form 36 (SF-36)	Post-intervention	MA
					Self-efficacy	Self-Efficacy for Energy Conserva- tion Questionnaire	Post-intervention	MA
García-Bur- guillo 2009 [144] Spain	RCT	Fatigue manage- ment	No informa- tion report- ed	14 (9/5)	Daily func- tioning	Fatigue Impact Scale (FIS)	Post-intervention	NS
					Quality of life	Short Form 36 (SF-36)	Post-intervention	NS
					Self-efficacy	Self-Efficacy for Energy Conserva- tion Questionnaire	Post-intervention	NS
Goverover 2018 [155] USA	RCT	Self-gener- ation learn- ing pro- gramme to improve	Memory tasks with- out self- generation learning in- structions	35 (19/16)	Daily func- tioning	Functional Behavior Profile (FBP)	Post-intervention	MA

Table 1. Overview of included studies, outcomes and synthesis *(Continued)*
learning and memory

					Quality of life	Functional Assessment of Multiple Sclerosis (FAMS)	Post-intervention	MA
					Self-management	Self-Regulation Skills Interview (SRSI)	Post-intervention	MA
					Mood (depression)	Chicago Multiscale Depression Inventory (CMDI)	Post-intervention	MA
					Mood (anxiety)	State-Trait Anxiety Inventory (STAI)	Post-intervention	MA
Hersche 2019 [145]	RCT	Fatigue management	Progressive muscle relaxation	47 (24/23)	Daily functioning	Modified Fatigue Impact Scale (MFIS)	Post-intervention Medium-term follow-up	MA
Switzerland					Quality of life	Short Form 36 (SF-36)	Post-intervention Medium-term follow-up	MA
					Self-efficacy	Self-Efficacy for Performing Energy Conservation Strategies Assessment (SEPECSA)	Post-intervention Medium-term follow-up	MA
Hynes 2024 [146, 194]	RCT	Cognitive Occupation-Based Programme	Usual care	110 (50/60)	Daily functioning	Modified Fatigue Impact Scale (MFIS)	Post-intervention Medium-term follow-up	MA
Ireland					Quality of life	Short Form 36 (SF-36)	Medium-term follow-up	MA
					Self-efficacy	General Self-Efficacy Scale (GSES)	Medium-term follow-up	MA
					Mood (depression)	General Health Questionnaire (GHQ-12)	Medium-term follow-up	MA

Table 1. Overview of included studies, outcomes and synthesis (Continued)

Jones 1996 [156] UK	CCT	Optimising ADL perfor- mance	No informa- tion provid- ed	62 (36/26)	Daily func- tioning	Northwick Park ADL Index (NPI)	Post-intervention	NS
Kalina 2018 [148] USA	RCT	Community Re-Integra- tion for So- cially Isolat- ed Patients	Usual care	91 (51/40)	Self-efficacy	Multiple Sclerosis Self-Efficacy (MSSE)	Post-intervention	MA
Kamm 2015 [160] Switzerland	RCT	Dexterity training	Theraband Training	39 (20/19)	Daily func- tioning	Dexterity-related ADL question- naire	Post-intervention	MA
Kessler 2024 [159] Canada	RCT	Occupa- tional Per- formance Coaching	Waiting list	30 (15/15)	Daily func- tioning	Canadian Occupational Perfor- mance Measure (COPM)	Post-intervention	MA
					Quality of life	Multiple Sclerosis Impact Scale (MSIS-29)	Post-intervention	MA
					Participation	Impact on Participation and Au- tonomy scale (IPA)	Post-intervention	MA
					Resilience	Connor-Davidson Resilience Scale-10 (CD-RISC10)	Post-intervention	MA
Kos 2016 [147] Belgium	RCT	Fatigue manage- ment	Relaxation programme	31 (17/14)	Daily func- tioning	Canadian Occupational Perfor- mance Measure (COPM)	Post-intervention	MA
					Quality of life	Short Form 36 (SF-36)	Post-intervention	MA
Mathiowetz 2005 [150] USA	RCT	Fatigue manage- ment	Waiting list	121 (62/69)	Daily func- tioning	Fatigue Impact Scale (FIS)	Post-intervention	NS
					Quality of life	Short Form 36 (SF-36)	Post-intervention	MA

Table 1. Overview of included studies, outcomes and synthesis (Continued)

					Self-efficacy	Self-Efficacy for Performing Energy Conservation Strategies Assessment (SEPECSA)	Post-intervention	NS
Ploew 2019 [151, 195]	RCT	Fatigue management	Social support intervention	139 (70/69)	Daily functioning	Fatigue Impact Scale (FIS)	Post-intervention Medium-term follow-up	MA
USA					Quality of life	Multiple Sclerosis Impact Scale (MSIS-29)	Post-intervention Medium-term follow-up	MA
					Participation	Godin Leisure-Time Exercise Questionnaire	Post-intervention Medium-term follow-up	MA
					Self-Efficacy	Self-Efficacy for Performing Energy Conservation Strategies Assessment (SEPECSA)	Post-intervention Medium-term follow-up	MA
Sánchez-Herera-Baeza 2022 [157]	CCT	Mental practice & motor manipulation skills training	Usual care	23 (13/10)	Daily functioning	Canadian Occupational Performance Measure (COPM)	Post-intervention Medium-term follow-up	NS
Spain								
Thomas 2013 [153, 161]	RCT	Fatigue management	Usual care	164 (84/80)	Quality of life	Multiple Sclerosis Impact Scale (MSIS-29)	Post-intervention Medium-term follow-up Long-term follow-up	MA
UK					Self-efficacy	Multiple Sclerosis Fatigue Self-Efficacy	Post-intervention Medium-term follow-up Long-term follow-up	MA

Table 1. Overview of included studies, outcomes and synthesis (Continued)

					Mood	Hospital Anxiety and Depression Scale (HADS)	Post-intervention Medium-term follow-up	MA
Vanage 2003 [152]	CCT	Fatigue management	Usual care	37 (21/16)	Daily functioning	Fatigue Impact Scale (FIS)	Post-intervention	NS
USA								

*In cross-over trials using waiting list control groups, only post-intervention assessment of the first arm is included in the analyses.

CCT: controlled clinical trial; MA: meta-analysis of standardised effect sizes; NEE: no effect estimate; NS: narrative synthesis; RCT: randomised controlled trial

Table 2. Results of studies not included in meta-analysis

Study	Study design	Intervention	Comparator	Time point	Outcome	Outcome measure	Results	Reason for exclusion from meta-analysis
Eyssen 2013	RCT	Client-centred occupational therapy	Usual care	Post-intervention, medium term	Daily functioning	Canadian Occupational Performance Measure (COPM)	Study authors narratively reported on COPM changes over time across the entire sample: "ITT and PP analyses on time effect showed significant improvements ($p < 0.05$) for...performance and satisfaction of meaningful activities..." [142, p. 1644].	Results were provided but not in time for analysis, due to technical issues reported by the study authors.
					Quality of life	Short Form 36 (SF-36)	Study authors reported differences in favour of the control group on two SF-36 subscales at post-intervention and one subscale at medium-term follow-up [142].	Results were provided but not in time for analysis, due to technical issues reported by the study authors.
					Participation	Impact on Participation and Autonomy (IPA)	Study authors reported no differences between intervention and control groups in the reported IPA subscales [142].	Results were provided but not in time for analysis, due to technical issues reported by the study authors.

Table 2. Results of studies not included in meta-analysis (Continued)

García-Bur- guillo 2009	RCT	Energy con- servation strategies	Usual care	Post-inter- vention	Daily func- tioning	Fatigue Im- pact Scale (FIS)	Study authors reported no signifi- cant changes over time in FIS in the intervention group [144].	Pre-post results for intervention group only
					Quality of life	Short Form 36 (SF-36)	Study authors reported no signifi- cant changes over time in SF-36 in the intervention group [144].	Pre-post results for intervention group only
					Self-efficacy	Self-Effica- cy for Energy Conservation Questionnaire	Study authors reported that "the SEPECSA questionnaire shows very high treatment adherence, with an average increase of 2.17 across all items". Translation [144, page 182]	Pre-post results for intervention group only
Mathiowetz 2005	RCT	Energy con- servation course	No interven- tion	Post-inter- vention	Daily func- tioning	Fatigue Im- pact Scale (FIS)	Study authors reported significant improvements on physical and social subscales of the FIS in intervention versus control group.	No results of total FIS scores
Jones 1996	CCT	Therapy in- tervention in reducing impairment and disabili- ty due to up- per limb and trunk ataxia	No interven- tion	Post-inter- vention	Daily func- tioning	Northwick Park ADL In- dex (NPI)	Intervention group improved ADL performance, except stair climbing ability, whereas controls did not ($P < 0.05$).	CCT with critical risk of bias
Sánchez- Her- rera-Baeza 2022	CCT	Mental prac- tice com- bined with training in motor ma- nipulation skills	Usual care	Post-inter- vention, medium term	Daily func- tioning	Canadian Oc- cupation- al Perfor- mance Mea- sure (COPM)	Intervention and control groups showed similar improvements in COPM at both time points.	CCT with critical risk of bias
Vanage 2003	CCT	Energy con- servation course	Usual care	Post-inter- vention	Daily func- tioning	Fatigue Im- pact Scale (FIS)	Intervention group showed improve- ment in FIS, whereas control group did not ($P < 0.05$).	CCT with critical risk of bias

CCT: controlled clinical trial; P: probability value; RCT: randomised controlled trial

Table 3. Results of risk of bias assessment of non-randomised clinical trials

Study	D1	D2	D3	D4	D5	D6	D7	Overall
Jones 1996	Critical	Serious	Moderate	Low	Critical	Moderate	Serious	Critical
Sánchez-Herrera-Baeza 2022	Serious	Serious	Moderate	Moderate	Serious	Moderate	Serious	Critical
Vanage 2003	Serious	Moderate	Serious	Serious	Serious	Moderate	Low	Critical

Based on ROBINS-I

Domain 1: risk of bias due to confounding

Domain 2: risk of bias in classification of interventions

Domain 3: risk of bias in selection of participants into the study (or into the analysis)

Domain 4: risk of bias due to deviations from intended interventions

Domain 5: risk of bias due to missing data

Domain 6: risk of bias arising from measurement of the outcome

Domain 7: risk of bias in selection of the reported result